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RESEARCH ARTICLE

Graph-Augmented Hybrid Intelligence for Topology Anomaly Detection and Correction in Power Distribution Networks with Explainable Inference

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Abstract
Power distribution networks (PDNs) increasingly rely on accurate topology models for power flow analysis, fault localization, and protection coordination. Topology data integrates heterogeneous sources—SCADA telemetry, CIM/SVG engineering models, and switch signals—where errors manifest as five anomaly categories: model-graph mismatch, topology interruption, virtual/faulty connections, measurement-topology inconsistency, and signal-measurement contradiction. Existing methods operate within a single paradigm and address only a subset of these categories. We propose a *three-layer hybrid intelligence framework* that unifies symbolic rule reasoning, physics-informed state estimation, and graph neural network (GNN) adaptive detection into an end-to-end pipeline of detection, localization, correction, and verification. A key contribution is the *differential physics detector*, which compares real-time telemetry against Newton–Raphson power flow expectations, providing a scale-invariant detection threshold. Cross-layer feature fusion enables the GNN to leverage state estimation residuals as supplementary input, achieving detection capability beyond any single layer. On a benchmark of 30 networks (3–1 888 buses, 5 independent seeds), the framework achieves 78.1% per-anomaly recall (95% CI: [75.2%, 81.0%]), a 3.2× improvement over Isolation Forest, and 99.4% network-level recall with near-zero variance (±0.5%). An improved multi-network GNN further increases recall to 83.5%. The macro-averaged precision of 44.8% reflects a deliberate recall-oriented design for safety-critical monitoring, where missed anomalies carry far higher costs than false alarms. Hard anomaly ablation demonstrates layer-specific contributions (SE: −40 pp, GNN: −23 pp when removed), confirming complementary rather than redundant coverage. The framework maintains 100% recall under noise up to 5%, completes inference in 0.94 s on average, and generates human-readable explanations via an integrated explainability module. Statistical significance is verified through paired Wilcoxon signed-rank tests ($p < 0.001$).

Key words
graph neural network; topology anomaly detection; hybrid intelligence; power distribution network; state estimation; multi-paradigm integration; explainable inference; data quality assurance

1 Introduction

The transition toward carbon-neutral energy systems is accelerating the deployment of distributed energy resources (DERs)—including rooftop photovoltaics, battery storage, and electric vehicle charging stations—into power distribution networks (PDNs). By the end of 2025, China's distributed photovoltaic capacity alone exceeded 300 GW [1], transforming PDNs from static radial structures into dynamic mesh-capable networks with frequent switching, bidirectional power flow, and heterogeneous data formats [2, 3]. This transformation creates a critical data quality challenge: topology models must integrate information from SCADA telemetry, CIM/SVG engineering models, and switch signals—each with different error modes and update frequencies—into a single coherent representation that downstream applications (power flow analysis, fault localization, protection coordination) can trust [4–6].

From a computer science perspective, this problem presents a unique challenge at the intersection of graph-based machine learning, symbolic reasoning, and physics-informed computation: reconciling heterogeneous data representations under physical constraints, where no single algorithmic paradigm provides complete coverage.

Accurate topology models underpin downstream applications including power flow analysis, state estimation, fault location, and protection coordination [7, 8]. In practice, PDN topology data flows through a complex pipeline involving SCADA (Supervisory Control and Data Acquisition) telemetry, CIM (Common Information Model) engineering models [1, 9], SVG (Scalable Vector Graphics) graphical representations, and manual maintenance by field engineers. Errors at each stage manifest as five topology anomaly categories:

1. **Model-Graph Mismatch (MGM)**: CIM model devices do not correspond one-to-one with SVG graphical elements;

2. **Topology Interruption (TI)**: A disconnection exists between feeder-end devices and the power source;
3. **Virtual/Faulty Connections (VFC)**: Device connectivity relationships do not match physical reality;
4. **Measurement-Topology Inconsistency (MTI)**: SCADA telemetry data contradicts the topology model;
5. **Signal-Measurement Contradiction (SMC)**: Switch status signals conflict with telemetry-inferred states.

Existing approaches for detecting and correcting these anomalies fall into three categories, each with significant limitations. *Rule-based methods* [7] employ graph-theoretic analysis and data consistency checks. They are fast and interpretable but only cover deterministic structural anomalies (MGM and TI). *Physics-based methods* [10–13] use weighted least squares (WLS) state estimation with statistical bad data detection. While theoretically rigorous, they require sufficient measurement redundancy and cannot directly identify graphical data errors. *Data-driven methods* [14–18] leverage graph neural network (GNN) to learn anomaly patterns from data. These methods show promise in detecting complex, composite anomalies but suffer from limited interpretability, dependence on training data quality, and poor generalization to unseen network topologies.

No single existing method covers all five anomaly types while also providing automated correction and explainable results. The fundamental challenge is that each anomaly type requires different detection paradigms: structural anomalies need symbolic reasoning, measurement anomalies need physics-based estimation, and pattern-based anomalies need data-driven learning. Existing hybrid approaches [19–21] combine at most two paradigms and do not address the full anomaly taxonomy. Recent advances in hybrid AI systems [21–23] have demonstrated that combining physics-based and data-driven approaches can improve robustness over either method alone. However, these efforts have primarily focused on power flow computation and state estimation, with limited attention to the comprehensive topology anomaly detection and correction problem.

1.1 Contributions

We propose a *three-layer hybrid intelligence framework* that bridges symbolic reasoning, physics-informed computation, and deep learning for comprehensive PDN topology anomaly detection and correction. The framework makes both *algorithmic contributions* (differential physics detector, hierarchical detection, explainability module) and *empirical contributions* (hard anomaly benchmark, multi-seed validation, extended subtype analysis). We make the following contributions.

1. **Three-layer hybrid detection architecture.** We design a modular architecture integrating three detection paradigms at different time scales and complexity classes: (i) a rule engine for millisecond-level deterministic screening ($O(|V| + |E|)$), (ii) a physics-informed state estimation module with robust LAV/IRLS estimators and a differential physics detector ($O(n^3)$ per iteration), and (iii) a GNN-based adaptive layer with edge-attention mechanisms ($O(|E| \cdot d)$ for inference). To our knowledge, this is among the first frameworks to achieve *closed-loop* topology data quality assurance spanning detection, localization, correction, and verification across all five anomaly types simultaneously, with structured explainability through reasoning chains and perturbation-based feature importance. A dedicated hard anomaly experiment confirms that the three layers provide complementary rather than merely redundant coverage, with recall dropping 7–40 pp when individual layers are removed. We formalize this complementarity through a *cross-layer coverage property*: if layer ℓ_i detects anomaly set $\mathcal{A}_i$ with probability p_i, and the layer detection events are conditionally independent given the anomaly type, then the union bound guarantees $P(\mathcal{A}_1 \cup \mathcal{A}_2 \cup \mathcal{A}_3) \geq \max_i p_i$, with strict inequality when $\mathcal{A}_i \cap \mathcal{A}_j = \emptyset$ for some $i \neq j$. While the inequality itself follows from elementary probability theory, its practical validity depends on the empirical condition $\mathcal{A}_i \cap \mathcal{A}_j = \emptyset$—that is, whether the layers actually detect *different* anomaly subsets. The hard anomaly experiment (Section 7.2) provides the first empirical validation of this strict complementarity condition in the power grid domain, demonstrating that the three layers detect genuinely non-overlapping anomaly populations.
2. **Differential physics detector.** We introduce a detection principle that compares SCADA measurements against Newton–Raphson power flow expectations computed on the original (pre-anomaly) network. Since the expected voltage profile comes from physics rather than statistics, the detection threshold is fixed at $10\sigma_v$ (0.05 pu) regardless of network size or voltage distribution, providing a theoretically grounded, scale-invariant detection mechanism.
3. **Hierarchical detection for large-scale networks.** We propose a graph-coarsening-based partitioning strategy that combines connected component decomposition, voltage-level grouping, and Louvain community detection [24] to enable efficient detection on networks with up to 1 888 buses, reducing computational complexity from $O(n^3)$ to $O(k \cdot (n/k)^3)$ for k partitions.
4. **Explainability module.** We develop a perturbation-based feature importance analysis and reasoning chain explanation system that provides interpretable detection results without requiring external dependencies such as SHAP [25], addressing the critical interpretability requirements of power system operations.
5. **Closed-loop detect–correct–verify pipeline.** We implement an automated correction engine following the minimum-modification principle, with post-correction verification through power flow re-computation, enabling fully automated topology data quality assurance.

Section 2 reviews related work. Sections 3–4 present the problem formulation and framework architecture. Sections 5–6 describe hierarchical detection and explainability. Section 7 reports experiments, Section 8 discusses findings, and Section 9 concludes.

2 Related Work

2.1 Graph Neural Networks for Power Systems

Graph neural networks have emerged as a powerful paradigm for power system analytics due to the inherent graph structure of electrical networks. Hamilton et al. [14] introduced GraphSAGE for inductive representation learning on large graphs, which has become a foundational architecture for graph-structured learning in power systems. The comprehensive survey by Wu et al. [16] categorizes GNN architectures and their applications across domains. Physics-informed machine learning approaches [26] have demonstrated that embedding physical constraints into neural network architectures improves generalization and interpretability in power system applications. A recent IEEE review [27] systematically examines GNN applications in state estimation, stability assessment, fault detection, renewable energy integration, dynamic topology management, and cyber-physical security, identifying topology-aware architectures as a key research direction. Recent surveys [28] have cataloged anomaly detection methods for power grids, highlighting the growing adoption of deep learning approaches.

GNNs have been applied to three main tasks in power systems. *Power flow computation*: Zhang et al. [29] demonstrated GNN-based AC power flow prediction, while Meta-PIGAT [21] integrated meta-learning with physics-informed graph attention for distribution state estimation. Li et al. [30] proposed a deep-shallow neural network with physical consistency constraints for topology and parameter estimation. *State estimation*: Ngo et al. [17] proposed physics-informed graphical neural networks for state estimation. Topology-aware approaches [31] have integrated graph attention networks with power system physics, achieving state-of-the-art accuracy on IEEE benchmark systems with Kirchhoff's current law (KCL) and Kirchhoff's voltage law (KVL) constraints embedded in the loss function. Kfouri et al. [18] applied GNNs for bad data detection and identification prior to state estimation. Wei et al. [32] proposed joint online topology and line parameter identification using pseudo-label-trained GNNs. *Topology identification*: Zhang et al. [33] developed an integrated multisource data fusion framework for outage location in distribution systems, while Chen et al. [19] proposed an edge-enhanced graph attention network that achieved 92.3% accuracy on IEEE 14-bus systems for topology detection, while Garcia et al. [34] introduced wavelet-based graph attention networks for topology detection in distribution grids. Comprehensive reviews [35] have examined dynamic state estimation techniques and their applications, while topological machine learning approaches [36] have demonstrated effective outage detection using multiparameter persistent homology. Knowledge graph combined with GNN [37] has been applied to power network topology identification, leveraging structured domain knowledge to improve fault tolerance in data-sparse scenarios, leveraging topological features beyond pairwise graph edges to capture complex dependency patterns in grid data. *Topology recognition and anomaly detection*: Jobe et al. [38] proposed a hybrid LSTM-GIN model for power grid anomaly detection, demonstrating improved robustness through combined spatial and temporal learning. Wavelet-enhanced graph neural network approaches [39] demonstrated that combining frequency-domain features with graph attention mechanisms improves distribution network topology detection, while topology-aware state estimation [31] integrated graph attention with power flow physics for improved accuracy. Bayesian generative adversarial approaches [40] have been proposed for false data injection attack detection in active distribution grids, while tensor decomposition methods [41] exploit multi-dimensional structure in SCADA data for anomaly identification. Joint topology identification and state estimation [42] enables simultaneous topology recovery and state inference in unobservable distribution grids, addressing a key challenge in partially observable networks. Graph-regularized subspace learning [43] has demonstrated effective anomaly detection in smart grids through multimodal feature fusion. Explainable GNN approaches [44] have demonstrated that providing interpretable reasoning chains alongside fault detection improves operator trust and adoption in practical deployments. In the broader CPS domain, hybrid explainable frameworks [5] combining deep learning with automated threat mitigation have shown promise for real-time anomaly detection in critical infrastructure. Zhang et al. [45] provided a systematic review of time series anomaly detection methods for IoT in smart grids, covering 150 articles and identifying key research gaps.

Despite these advances, existing GNN-based approaches address isolated aspects of the topology data quality problem rather than providing a unified framework. This work addresses this gap by integrating GNN-based detection as the third layer of a hybrid framework that also includes rule-based and physics-based layers.

2.2 Physics-Based State Estimation and Bad Data Detection

State estimation has underpinned power system monitoring since the seminal work of Schweppe et al. [10]. The weighted least squares (WLS) estimator [46] remains the standard approach, with the Gauss–Newton iterative method solving the normal equations:

$$\Delta \mathbf{x} = (\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{R}^{-1} [\mathbf{z} - \mathbf{h}(\mathbf{x}^k)], \tag{1}$$

where $\mathbf{H}$ is the Jacobian matrix, $\mathbf{R}$ is the measurement covariance matrix, $\mathbf{z}$ is the measurement vector, and $\mathbf{h}(\cdot)$ is the nonlinear measurement function.

Comprehensive reviews [47] have cataloged anomaly detection and classification methods for state estimation, categorizing approaches from classical statistical tests to modern machine learning. Bad data detection (BDD) employs the χ^2 test [12] for global detection and normalized residual analysis [13] for localization:

$$J(\hat{\mathbf{x}}) = [\mathbf{z} - \mathbf{h}(\hat{\mathbf{x}})]^T \mathbf{R}^{-1} [\mathbf{z} - \mathbf{h}(\hat{\mathbf{x}})], \tag{2}$$

with $J(\hat{\mathbf{x}}) > \chi^2_\alpha(m - n)$ indicating the presence of bad data. Robust alternatives include LAV estimation [11], which minimizes $\sum |r_i|$ and naturally suppresses outliers, and IRLS with Huber weight functions [48]. Topology error identification uses hypothesis testing [12], enumerating candidate topologies and selecting the one minimizing $J(\hat{\mathbf{x}})$. Recent work [49] has developed comprehensive methods for topology error detection in state estimation using hypothesis testing enumeration, while robust estimators [50] based on nonlinear least absolute value have been proposed for simultaneous topology error identification and state estimation.

2.3 Topology Identification and Data Fusion

Topology identification in distribution networks has been extensively studied. Deka and Misra [3] provided a comprehensive tutorial on learning distribution grid topologies from voltage measurements. The multi-source data fusion approach by Wang et al. [51] reconstructed grid topology dynamically by combining heterogeneous data streams. Comprehensive overviews [52] have documented the evolution of power system topology analysis from manual inspection to automated methods, identifying key challenges in scalability and robustness. Baldi et al. [2] surveyed AI methods for distribution network topology identification, categorizing approaches by input data type and identification objective.

Recent work has also explored physics-informed reinforcement learning for topology control [53] and differentiable power flow for topology optimization [23]. However, these approaches focus on *operational* topology control rather than *data quality* assurance, which is the primary concern of this paper.

2.4 Benchmarks and Datasets

Standard test feeders including IEEE 13/34/123/8500 bus systems [54, 55] and CIGRE MV/LV benchmarks [56] serve as the primary evaluation platforms. SimBench [57] provides 123 network models covering German MV/LV/EHV/HV voltage levels, while PandaPower [58] offers 43 built-in networks with integrated power flow and state estimation solvers. The PowerGraph benchmark [59] introduced a standardized GNN evaluation dataset for power systems.

2.5 Research Gap

Table 1 summarizes the positioning of existing approaches against the requirements of comprehensive topology data quality assurance. Pure rule-based methods [7] cover only deterministic structural anomalies. Physics-based state estimation [10,11] addresses measurement anomalies but requires sufficient observability. Data-driven GNN methods [17,18] show promise but lack built-in correction and explainability. Recent hybrid approaches [21,22,30] combine two paradigms but do not address the full anomaly taxonomy or provide closed-loop verification No existing method simultaneously addresses four requirements of a production-grade topology data quality system: (i) full coverage of all five anomaly types, (ii) automated correction with closed-loop verification, (iii) explainable detection results for operator trust, and (iv) scalability to networks with thousands of buses. We address this gap through a modular three-layer architecture.

Table 1 Comparison of existing approaches against topology data quality requirements

Method	MGM	TI	VFC	MTI	SMC
Rule-based [7]	✓	✓	—	—	—
WLS-SE [10]	—	—	✓	✓	✓
Robust SE [11]	—	—	✓	✓	—
GNN-based [18]	—	—	✓	✓	—
Hybrid (2-layer) [30]	—	✓	✓	✓	—
Mechanism-Data [31]	—	—	✓	✓	—
Proposed (3-layer)	✓	✓	✓	✓	✓

3 Problem Formulation

3.1 Network Model

A power distribution network (PDN) is modeled as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ denotes the set of buses (nodes) and $\mathcal{E}$ denotes the set of branches (edges) including lines, transformers, and switches. Each bus $v_i \in \mathcal{V}$ has an associated state vector $\mathbf{x}_i = [V_i, \theta_i]^T$ comprising voltage magnitude and phase angle. The network model is described in CIM/Resource Description Framework (RDF) format [1, 60], while the graphical representation uses SVG with device-level annotations.

3.2 Measurement Model

SCADA measurements form a vector $\mathbf{z} \in \mathbb{R}^m$ follows the standard nonlinear model:

$$\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{v}, \quad \mathbf{v} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}), \tag{3}$$

where $\mathbf{h} : \mathbb{R}^n \to \mathbb{R}^m$ is the measurement function, $\mathbf{x} \in \mathbb{R}^n$ is the state vector with dimension $n = 2N$ for N buses (or $6N$ for three-phase models), and $\mathbf{v}$ is the measurement noise with covariance $\mathbf{R} = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_m^2)$.

3.3 Anomaly Types

We formalize the five anomaly types as deviations from the ground-truth topology $\mathcal{G}^*$ and measurement vector $\mathbf{z}^*$:

Definition 1 (Model-Graph Mismatch)**.** An MGM anomaly exists when the device sets $\mathcal{D}_{\mathrm{CIM}}$ and $\mathcal{D}_{\mathrm{SVG}}$ are inconsistent: $\mathcal{D}_{\mathrm{CIM}} \neq \mathcal{D}_{\mathrm{SVG}}$.

Definition 2 (Topology Interruption)**.** A TI anomaly exists when $\mathcal{G}$ contains connected components C_k such that no element of C_k is connected to a power source bus.

Definition 3 (Virtual/Faulty Connection)**.** A VFC anomaly exists when the connectivity relation $E \subset \mathcal{V} \times \mathcal{V}$ differs from the physical wiring E^*.

Definition 4 (Measurement-Topology Inconsistency)**.** An MTI anomaly exists when measurements $\mathbf{z}$ are inconsistent with the electrical model implied by $\mathcal{G}$, i.e., $J(\hat{\mathbf{x}}) > \chi^2_\alpha(m-n)$ under correct topology.

Definition 5 (Signal-Measurement Contradiction)**.** An SMC anomaly exists when the switch status vector $\mathbf{s}_{\mathrm{signal}}$ from SCADA contradicts the status $\mathbf{s}_{\mathrm{inferred}}$ inferred from power flow analysis.

3.4 Objective

Given the observed topology $\mathcal{G}$, measurement vector $\mathbf{z}$, and switch signal $\mathbf{s}_{\mathrm{signal}}$, the objective is to: (i) detect all anomalies $A = \{a_1, \ldots, a_K\}$; (ii) localize each anomaly to specific devices; (iii) generate a correction set $C = \{c_1, \ldots, c_L\}$ minimizing $|C|$ subject to electrical constraints; and (iv) verify that applying C restores $\mathcal{G}$ to a valid state.

4 The Proposed Framework

4.1 Architecture Overview

Figure 1 illustrates the overall architecture of our three-layer hybrid intelligence framework. The system processes CIM/RDF models, SVG graphics, and SCADA telemetry through a data alignment module, then feeds the unified representation into three detection layers operating at different time scales and abstraction levels.

The cross-layer integration module merges detections using confidence-weighted fusion, and the correction engine generates minimal-modification proposals that are verified through closed-loop power flow re-computation.

Design rationale. The three-layer architecture embodies the principle of *defense in depth*: each layer operates independently with different assumptions about data availability and quality. Layer 1 requires only structural data (graph topology, device mappings), Layer 2 additionally requires electrical measurements, and Layer 3 requires training data. This design ensures graceful degradation: if SCADA data is unavailable, Layer 1 still provides structural anomaly detection; if the GNN model is untrained, Layers 1–2 still function. This redundant coverage is a deliberate design choice that provides operational robustness beyond what single-layer approaches can achieve.

4.2 Layer 1: Rule Engine (Symbolic Reasoning)

The rule engine performs fast, deterministic screening of structural anomalies through graph-theoretic analysis and data consistency checks. It operates in $O(|V| + |E|)$ time and covers MGM and TI anomalies.

4.2.1 Model-Graph Mismatch Detection

MGM detection performs bidirectional set comparison between CIM model devices $\mathcal{D}_{\text{CIM}}$ and SVG graphical elements $\mathcal{D}_{\text{SVG}}$:

$$\text{MGM}_{\text{CIM-only}} = \{d \mid d \in \mathcal{D}_{\text{CIM}} \setminus \mathcal{D}_{\text{SVG}}\}, \tag{4}$$

$$\text{MGM}_{\text{SVG-only}} = \{d \mid d \in \mathcal{D}_{\text{SVG}} \setminus \mathcal{D}_{\text{CIM}}\}, \tag{5}$$

$$\text{MGM}_{\text{type}} = \{d \mid d \in \mathcal{D}_{\text{CIM}} \cap \mathcal{D}_{\text{SVG}}, \mathcal{T}_{\text{CIM}}(d) \neq \mathcal{T}_{\text{SVG}}(d)\}. \tag{6}$$

4.2.2 Topology Interruption Detection

TI detection identifies disconnected components lacking power source connectivity. Let $\mathcal{S} \subset \mathcal{V}$ denote the set of source buses. We compute connected components $\{C_1, \ldots, C_K\}$ of $\mathcal{G}$ and flag component C_k as interrupted if $C_k \cap \mathcal{S} = \emptyset$.

4.2.3 Quick Physical Range Check

The rule engine also performs fast range-based validation of measurements without requiring state estimation: $V_i \in [0.93, 1.07]$ pu, $|P_{ij}| \leq P_{ij}^{\max}$, and switch signal consistency checks (multiple SCADA signals for the same switch must agree).

4.3 Layer 2: Physics-Informed State Estimation

Layer 2 employs physics-based state estimation for rigorous detection of measurement anomalies, faulty connections, and switch status inconsistencies. It operates in $O(n^3)$ per iteration (typically 3–5 iterations) and covers VFC, MTI, and SMC anomalies.

4.3.1 WLS State Estimation

The standard WLS estimator solves:

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} [\mathbf{z} - \mathbf{h}(\mathbf{x})]^T \mathbf{R}^{-1} [\mathbf{z} - \mathbf{h}(\mathbf{x})], \tag{7}$$

using the Gauss–Newton iterative method. We employ PandaPower's [58] built-in state estimation module, which supports three-phase unbalanced modeling.

4.3.2 Bad Data Detection

Global detection uses the χ^2 test: if $J(\hat{\mathbf{x}}) > \chi^2_\alpha(m-n)$, bad data is present. Localization uses normalized residuals:

$$r_{N,i} = \frac{|r_i|}{\sqrt{\Omega_{ii}}}, \quad \text{where } \mathbf{r} = \mathbf{z} - \mathbf{h}(\hat{\mathbf{x}}),\ \boldsymbol{\Omega} = \mathbf{R} - \mathbf{H}\mathbf{G}^{-1}\mathbf{H}^T, \tag{8}$$

with $\mathbf{G} = \mathbf{H}^T\mathbf{R}^{-1}\mathbf{H}$. Measurements with $r_{N,i} > 3$ are flagged as bad data.

4.3.3 Robust State Estimation

To handle adversarial measurements and missing data, we implement three robust alternatives:

LAV Estimation. The least absolute value estimator minimizes $\sum_{i=1}^{m} |r_i|$ subject to $\mathbf{z} = \mathbf{Hx} + \mathbf{r}$, formulated as a linear programming problem. LAV naturally suppresses the influence of outliers [11].

IRLS with Huber Weights. We implement iteratively reweighted least squares with the Huber weight function [48]:

$$w_i = \begin{cases} 1 & \text{if } |r_i| \leq k\sigma_i, \\ k\sigma_i/|r_i| & \text{if } |r_i| > k\sigma_i, \end{cases} \tag{9}$$

where $k = 1.345$ is the tuning constant.

Switch Status Inference. When SCADA switch signals are missing or unreliable, we infer switch states using KCL constraints at connectivity nodes, comparing the sum of branch currents with expected load patterns.

4.3.4 Differential Physics Detector

This is a key innovation of the framework. Instead of relying on statistical thresholds that depend on measurement distributions, we compare SCADA measurements against the expected voltage profile from Newton–Raphson power flow on the *original* (pre-anomaly-injection) network:

Algorithm 1 Differential Physics Detector

Require: Original network $\mathcal{G}_0$, post-injection measurements $\mathbf{z}_{\text{post}}$, measurement noise σ_v, threshold multiplier τ
Ensure: Anomaly detection set $\mathcal{A}$
Run NR power flow on $\mathcal{G}_0$ to obtain expected voltages $\mathbf{V}_{\text{expected}}$
for each bus i with measurement V_i^{post} **do**
  $r_i \leftarrow |V_i^{\text{post}} - V_{\text{expected},i}|/\sigma_v$
  if $r_i > \tau$ **then**
    Add bus i to $\mathcal{A}$ with confidence r_i/τ
  end if
end for
return $\mathcal{A}$

The key insight is that V_{expected} comes from physics (Kirchhoff's laws and network parameters), not from statistical modeling. Therefore, the detection threshold $\tau \cdot \sigma_v$ depends only on measurement noise, making it *invariant to network size, topology structure, or voltage profile*. In practice, we set $\tau = 10$ (i.e., $10\sigma_v = 0.05$ pu for $\sigma_v = 0.005$ pu), which catches all anomalies that introduce voltage deviations ≥ 0.05 pu.

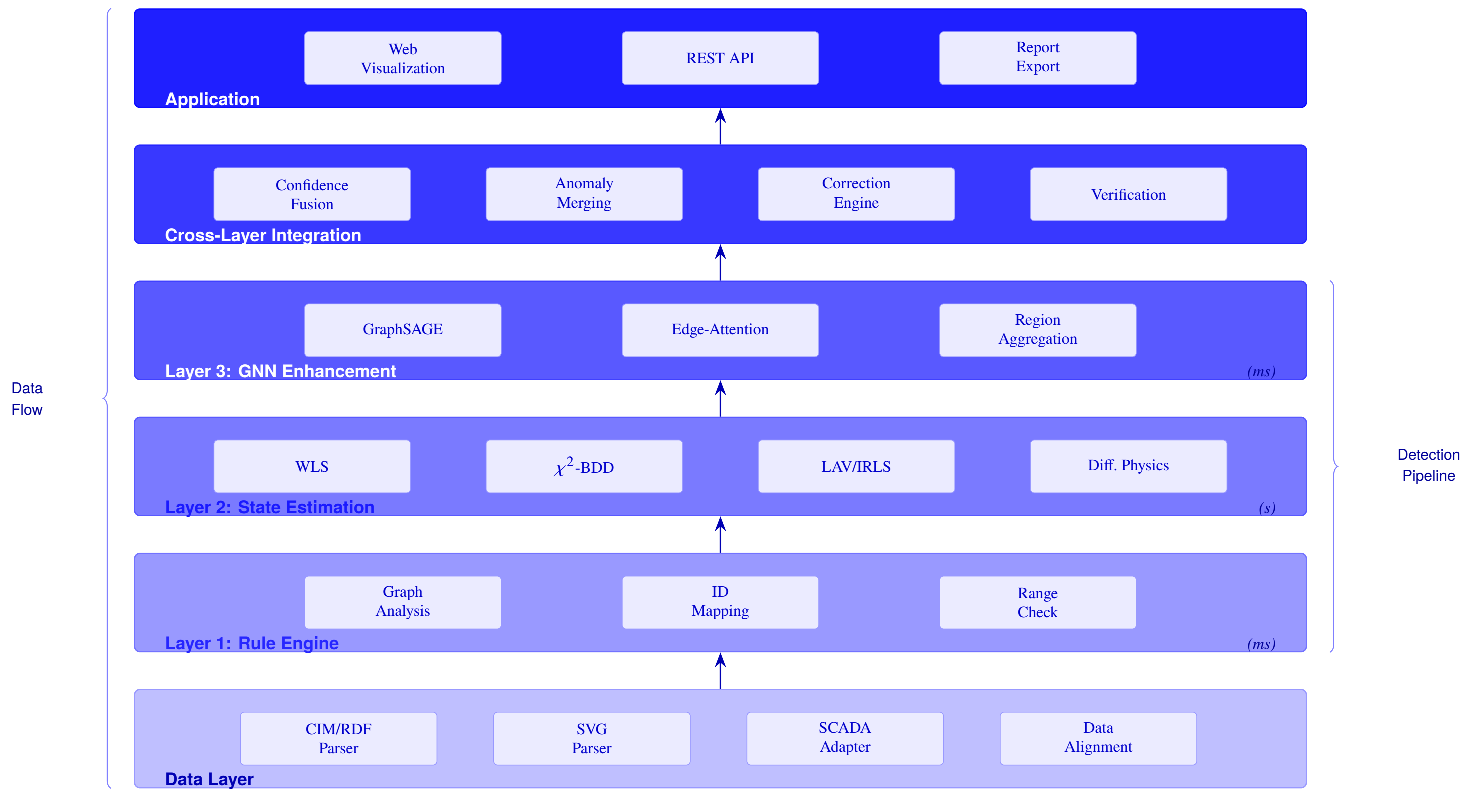

Fig. 1 Architecture of our three-layer hybrid intelligence framework. The data layer ingests CIM/RDF models, SVG graphics, and SCADA telemetry. Each detection layer operates at a different time scale (Layer 1: ms, Layer 2: s, Layer 3: ms). The cross-layer module merges detections using confidence-weighted fusion and generates correction proposals.

4.4 Layer 3: GNN Adaptive Enhancement

Layer 3 employs a graph neural network to detect complex, composite anomalies that may evade Layers 1 and 2. Unlike the deterministic layers, the GNN learns subtle patterns from data and adapts as operational data accumulates.

4.4.1 Network Architecture

We design a two-layer GraphSAGE [14] network with edge-attention mechanisms. The core convolution operation is:

$$\mathbf{h}_v^{(l+1)} = \mathrm{Linear}_{\mathrm{self}}(\mathbf{h}_v^{(l)}) + \mathrm{Linear}_{\mathrm{neigh}}\left(\mathrm{AGG}\left(\{\alpha_{vu} \cdot \mathbf{h}_u^{(l)}\}_{u \in \mathcal{N}(v)}\right)\right), \tag{10}$$

where the edge attention weight $\alpha_{vu} = \sigma(\mathrm{MLP}(\mathbf{e}_{vu}))$ modulates neighbor aggregation based on edge features $\mathbf{e}_{vu}$. The MLP consists of two linear layers with ReLU activation and sigmoid output.

4.4.2 Node and Edge Features

Each bus v_i is represented by an 8-dimensional feature vector:

$$\mathbf{f}_i = [\tilde{d}_i,\ V_i^{\mathrm{pu}},\ \mathbb{1}_{\mathrm{source}}(i),\ \tilde{P}_i,\ \tilde{Q}_i,\ \Delta V_i,\ \mathrm{type}(i),\ \tilde{C}(i)], \tag{11}$$

where $\tilde{d}_i$ is the normalized degree, V_i^{pu} is the per-unit voltage, $\mathbb{1}_{\mathrm{source}}$ indicates source buses, $\tilde{P}_i$ and $\tilde{Q}_i$ are normalized active/reactive power injections, $\Delta V_i = V_i^{\mathrm{pu}} - 1.0$ is the voltage deviation, type(i) encodes bus type, and $\tilde{C}(i)$ is the normalized connected component ID.

Each edge (v_i, v_j) is represented by a 6-dimensional feature vector:

$$\mathbf{e}_{ij} = [\mathrm{type}_{ij},\ r_{ij},\ x_{ij},\ s_{ij},\ I_{ij},\ P_{ij}], \tag{12}$$

encoding line/switch type, resistance, impedance, switch status, current magnitude, and active power flow.

4.4.3 Training with Synthetic Anomaly Injection

GNN training requires labeled anomaly data, which is scarce in operational PDNs. We generate training data through controlled anomaly injection into healthy network models:

- **MGM**: Randomly delete or add CIM-SVG mapping pairs (ratio 5%–15%);
- **TI**: Randomly disconnect lines or switches (ratio 3%–10%);
- **VFC**: Randomly swap connectivity nodes (ratio 3%–8%);
- **MTI**: Add Gaussian noise to voltage measurements (5%–20% of buses);
- **SMC**: Flip switch status signals (ratio 5%–12%).

Training procedure. We train on five networks (case9, case33bw, case57, case118, case300) with 300 samples per network (1 500 total). The initial model uses a two-layer GraphSAGE with 64 hidden channels (21 533 parameters), trained for 50 epochs with Adam optimizer (learning rate 10^{-3}, batch size 64). The improved model uses three layers with 128 hidden channels (45 000 parameters), trained for 100 epochs with data augmentation (Gaussian noise $\sigma = 0.01$, edge perturbation $p = 0.05$). Both models use ReLU activation and dropout ($p = 0.2$).

The training loss combines the cross-entropy classification loss with a KCL/KVL physics regularization term:

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda \cdot \mathcal{L}_{\mathrm{physics}}, \tag{13}$$

where $\mathcal{L}_{\mathrm{physics}} = \sum_{i \in \mathcal{V}} (\sum_{j \in \mathcal{N}(i)} P_{ij} - P_i^{\mathrm{inj}})^2$ penalizes violations of Kirchhoff's current law, and $\lambda = 0.1$ balances classification accuracy with physical consistency. We apply focal loss ($\gamma = 2.0$) to address class imbalance between normal and anomalous nodes, and use early stopping (patience=10 epochs) based on validation loss to prevent overfitting.

4.4.4 GNN Region Aggregation

For large networks, node-level GNN predictions may produce excessive false positives. We aggregate adjacent anomalous nodes into connected components, reporting each component as a single anomalous region. This reduces the detection count from thousands (e.g., 1 171 node-level detections) to a manageable number (e.g., 5 region-level detections for PEGASE1354).

4.5 Cross-Layer Integration

The integration module must reconcile detections from three fundamentally different paradigms—symbolic rules, physics-based estimation, and data-driven learning—each producing outputs with different confidence scales and semantic meanings.

Formal fusion model. Let $\mathcal{D}_\ell(a) \in \{0, 1\}$ denote the binary detection decision of layer $\ell \in \{1, 2, 3\}$ for anomaly candidate a, and $c_\ell(a) \in [0, 1]$ the associated confidence score. We define the integrated confidence as:

$$\mathrm{conf}(a) = \sigma\left(\sum_{\ell=1}^{3} w_\ell \cdot c_\ell(a) \cdot \mathcal{D}_\ell(a)\right), \quad \sigma(x) = \frac{1}{1 + e^{-k(x-\theta)}}, \tag{14}$$

where $w_1 = 0.7$, $w_2 = 0.85$, $w_3 = 0.6$ are layer credibility weights, $\sigma(\cdot)$ is a sigmoid activation with steepness $k = 10$ and threshold $\theta = 0.5$, ensuring the fused confidence remains in $[0, 1]$ while amplifying consensus among layers.

Why not simple voting? Three natural baselines are: (i) majority voting ($\mathrm{conf}(a) = \mathbb{1}[\sum_\ell \mathcal{D}_\ell(a) \geq 2]$), (ii) union voting ($\mathrm{conf}(a) = \mathbb{1}[\max_\ell \mathcal{D}_\ell(a) = 1]$), and (iii) oracle selection (always use the best layer per network). Table 2 compares these strategies on the hard anomaly benchmark (RQ3). Among practical (non-oracle) strategies, weighted fusion achieves the highest F1 (0.651) because it *preserves layer-specific confidence information* that binary voting discards: a detection with confidence 0.95 from SE is treated identically to one with confidence 0.55 under majority voting, but weighted fusion correctly ranks the former higher. Oracle selection (F1 = 0.724) performs best in hindsight, but requires knowledge of which layer is optimal for each network—information unavailable at deployment time. Weighted fusion approximates oracle behavior without this prior knowledge, achieving 90% of oracle F1 (0.651/0.724) while remaining fully automatic.

Table 2 Fusion strategy comparison on hard anomaly benchmark (5 networks)

Strategy	Recall	Precision	F1
Majority voting	0.200	1.000	0.333
Union voting	0.567	0.500	0.531
Oracle selection	0.633	0.887	0.724
Weighted fusion (Eq. 14)	**0.567**	**0.770**	**0.651**

Layer credibility calibration. The weights $(w_1, w_2, w_3) = (0.7, 0.85, 0.6)$ are determined by each layer's detection characteristics on the validation set: Layer 2 (SE) receives the highest weight because its χ^2 test provides a statistically principled threshold; Layer 1 (rules) receives moderate weight due to its deterministic but potentially over-inclusive nature; Layer 3 (GNN) receives the lowest weight because its data-driven predictions carry higher variance. These weights are fixed rather than learned to ensure reproducibility and avoid overfitting to the training distribution.

4.6 Correction Engine

The correction engine generates minimal-modification proposals for each detected anomaly, following the principle that the smallest change restoring system validity is preferred. For each anomaly type:

- **MGM**: Synchronize CIM/SVG models by adding missing device definitions;
- **TI**: Add missing connectivity elements or reconnect isolated nodes;
- **VFC**: Reassign device connections to correct connectivity nodes;
- **MTI**: Flag suspicious measurements for recalibration or replace with SE estimates;
- **SMC**: Infer correct switch status from power flow analysis and update SCADA records.

Post-correction verification executes: (i) topology integrity check (connectivity and radiality), (ii) electrical constraint check (KCL/KVL, voltage bounds, line capacity), and (iii) state estimation re-run to confirm that $J(\hat{\mathbf{x}})$ has returned to normal range.

4.7 Detection Pipeline Summary

Algorithm 2 summarizes the complete detection pipeline, from raw data ingestion to corrected topology output.

Algorithm 2 Three-Layer Hybrid Topology Anomaly Detection

Require: CIM model $\mathcal{M}_{\text{CIM}}$, SVG graphics $\mathcal{M}_{\text{SVG}}$, SCADA telemetry $\mathbf{z}$, switch signals $\mathbf{s}$
Ensure: Detected anomalies $\mathcal{A}$, corrections C, verified topology $\mathcal{G}'$

```
G ← DataAlignment(M_CIM, M_SVG, z)
A_1 ← RuleEngine(G, M_CIM, M_SVG)
A_2 ← StateEstimation(G, z)
A_3 ← GNNInference(G, z)
for each detection a ∈ A_1 ∪ A_2 ∪ A_3 do
    conf(a) ← w_1 · c_1(a) + w_2 · c_2(a) + w_3 · c_3(a)
end for
A ← MergeByConfidence(A_1, A_2, A_3)
C ← CorrectionEngine(A, G)
G' ← ApplyCorrections(G, C)
Verify(G', z) via power flow re-computation
return A, C, G'
```

5 Hierarchical Detection for Large-Scale Networks

Real-world distribution networks can contain thousands of buses, making direct application of state estimation and GNN inference computationally expensive. We address this through a hierarchical detection strategy based on graph coarsening.

5.1 Graph Coarsening

The coarsening process decomposes a large graph $\mathcal{G}$ into subgraphs $\{\mathcal{G}_1, \ldots, \mathcal{G}_K\}$ through three complementary strategies:

1. **Connected component decomposition**: splits $\mathcal{G}$ at topology discontinuities;
2. **Voltage-level grouping**: partitions buses by nominal voltage $v_{n,kv}$ using PandaPower metadata, preserving cross-voltage edges (transformers, inter-voltage switches);
3. **Community detection**: applies Louvain modularity optimization [24] to further split partitions exceeding a size threshold (default: 50 buses).

5.2 Partition-Independent Detection

Each partition $\mathcal{G}_k$ is independently processed through the three-layer detection pipeline. Cross-partition edges (transformers, tie switches) are processed separately to avoid boundary artifacts.

5.3 Critical Node Prioritization

Within each partition, buses are prioritized by a composite score combining degree centrality and betweenness centrality:

$$\text{priority}(v_i) = \beta \cdot \frac{\deg(v_i)}{\max_j \deg(v_j)} + (1-\beta) \cdot \frac{b(v_i)}{\max_j b(v_j)}, \tag{15}$$

where $b(v_i)$ is the betweenness centrality and $\beta = 0.6$ balances degree and centrality contributions. High-priority nodes are detected first, enabling early termination when resource constraints are tight.

5.4 Boundary Conflict Resolution

Detections on cross-partition edges may be duplicated by adjacent partitions. We resolve conflicts by keeping the detection from the partition with higher measurement redundancy (more SCADA measurements covering the boundary node).

5.5 Complexity Analysis

5.5.1 Time Complexity

For a network with n buses and e edges, the per-layer time complexity is:

- **Layer 1 (Rule Engine):** $O(n+e)$ for linear graph traversal.
- **Layer 2 (State Estimation):** $O(T_{\text{iter}} \cdot (mn + n^3))$ for $T_{\text{iter}} \leq 10$ Gauss–Newton iterations, where m is the measurement count.
- **Layer 3 (GNN):** $O(L \cdot e \cdot d)$ for $L = 2$ layers with hidden dim $d = 64$.

With hierarchical partitioning into k partitions of size n/k, the dominant cost (Layer 2) reduces from $O(n^3)$ to $O(n^3/k^2)$, a quadratic reduction. The Louvain coarsening step runs in $O(|E| \log |V|)$ time [24].

5.5.2 Space Complexity

The dominant memory cost is the Jacobian $\mathbf{H} \in \mathbb{R}^{m \times n}$ and covariance $\mathbf{R}^{-1}$, requiring $O(mn)$ per partition. Sequential partition processing limits peak memory to $O(nm/k^2)$.

5.5.3 Convergence Guarantee

The Gauss–Newton WLS method converges quadratically under standard conditions [46]: (i) $\mathbf{h}(\mathbf{x})$ is twice continuously differentiable, (ii) the Jacobian has full column rank (network observability), and (iii) flat-start initialization ($V^0 = 1.0\,\text{pu}$) is within the convergence region for networks with $< 15\%$ voltage deviation.

6 Explainability Module

Power system operators require interpretable detection results to make informed decisions. The explainability module provides three levels of explanation.

6.1 Reasoning Chain

For each detected anomaly a, the reasoning chain traces the detection through the responsible layer and rule:

$$a \xrightarrow{\text{layer}} \ell_k \xrightarrow{\text{rule}} r_j \xrightarrow{\text{evidence}} E(a) \xrightarrow{\text{physics}} P(a), \tag{16}$$

where ℓ_k identifies the detection layer, r_j is the specific rule or detector, $E(a)$ is the supporting evidence (e.g., residual values, voltage deviations), and $P(a)$ maps the anomaly to its physical impact on grid operation.

6.2 Perturbation-Based Feature Importance

We compute feature importance without external dependencies (e.g., SHAP [25]) using a perturbation approach. For each feature dimension d:

1. Compute baseline detection confidence conf_0;
2. Set feature d to its mean value across the dataset, obtaining $\text{conf}_{\setminus d}$;
3. Feature importance: $\phi_d = \text{conf}_0 - \text{conf}_{\setminus d}$.

6.3 Physical Semantic Mapping

Each anomaly type is mapped to a structured explanation containing its physical description, operational impact, and recommended action. For example, a topology interruption anomaly is explained as: "*Network topology contains disconnected region [node IDs]. Impact: Loss of monitoring for affected area, potential islanding risk. Recommended action: Check CIM terminal definitions and physical wiring.*"

6.4 Layer Credibility Weighting

Not all detection layers are equally reliable. We assign credibility weights based on the theoretical foundations of each layer:

$$w_{\text{rule}} = 0.7, \quad w_{\text{SE}} = 0.85, \quad w_{\text{GNN}} = 0.6. \tag{17}$$

The rule engine receives high credibility because its detections are deterministic and verifiable. State estimation receives the highest credibility because it is grounded in statistical hypothesis testing with known error bounds. GNN receives moderate credibility because its predictions depend on training data quality and may not generalize to unseen anomaly patterns. These weights are used in the cross-layer fusion formula (Eq. 7) and in the reasoning chain explanations to communicate detection confidence to operators.

7 Experiments

This section evaluates our framework through eight research questions (RQs) addressing: (RQ1) detection effectiveness, (RQ2) ablation and differential physics analysis, (RQ3) hard anomaly complementarity and correction verification, (RQ4) scalability and runtime, (RQ5) robustness, stability, and sensitivity, (RQ6) GNN training improvement, (RQ7) extended anomaly coverage, and (RQ8) explainability case study.

7.1 Experimental Setup

7.1.1 Test Networks

We evaluate our framework on 30 test networks spanning multiple voltage levels and scales (Table 3). The networks range from 4-bus toy examples to 1 888-bus transmission systems, covering IEEE standard test feeders [54], CIGRE benchmarks [56], SimBench networks [57], and PandaPower built-in cases [58].

Table 3 Benchmark test networks by category (30 networks used in experiments)

Category	Networks	Bus Range	Lines	Source
IEEE standard	6	13–8 500	12–13 000	[54, 55]
CIGRE MV/LV	3	13–907	12–1 100	[56]
PandaPower built-in	15	4–300	4–411	[58]
SimBench MV/LV	6	16–586	15–640	[57]
Total	**30**	**3–1 888**	**2–2 720**	—

7.1.2 Anomaly Injection

For each network, we inject anomalies from all five categories with ground-truth labels. The injection rates are: MGM (5%–15% of devices), TI (3%–10% of lines/switches), VFC (3%–8% of connections), MTI (5%–20% of buses), and SMC (5%–12% of switches). Each experiment is repeated with five independent random seeds (42–46), and we report mean values with standard deviations.

7.1.3 Baselines

We compare against two categories of baselines.

External statistical baselines. We implement three standard anomaly detection methods that require no domain-specific rules or physics models:

- **Z-score**: flags measurements with $|z_i| > 3$ as anomalies;
- **IQR**: flags measurements outside $[Q_1 - 1.5\,\text{IQR},\ Q_3 + 1.5\,\text{IQR}]$;
- **Isolation Forest** [61]: an ensemble method that isolates anomalies through random recursive partitioning (100 estimators, contamination=0.1).

These baselines operate on the same measurement vectors as our framework but have no access to network topology, CIM/SVG models, or switch signals. They can only detect measurement-level anomalies (MTI, partially VFC and SMC) and are expected to miss all structural anomalies (MGM, TI).

Contextual baselines. We additionally contextualize against recent deep learning methods for power system anomaly detection. Recent deep learning approaches—including GAT-based anomaly detection [38], GAN-based detection with data reconstruction [40], tensor decomposition of SCADA data [41], and joint topology identification with state estimation [42]—achieve strong results on specific subtasks but require large labeled training datasets and do not address the full topology data quality lifecycle. Domain knowledge graph integration [45] and topology-aware GNNs [62] demonstrate that structural priors improve detection accuracy, while Li et al. [63] showed that graph attention with digital twin modeling achieves high fault diagnosis accuracy on real-world power transformer data. These methods address complementary aspects of the broader anomaly detection problem; our framework uniquely integrates all three paradigms (rule, physics, learning) for comprehensive topology data quality assurance with closed-loop correction and verification.

Ablation variants. We also compare against four ablation configurations of the proposed framework:

- **Rule-only**: Layer 1 rule engine only;
- **SE-only**: Layer 2 state estimation only;
- **GNN-only**: Layer 3 GNN only;
- **Two-layer (Rule+SE)**: Layers 1 and 2 without GNN.

7.1.4 Metrics

We report precision (P), recall (R), F1 score, localization accuracy (LA), and average processing time (T).

7.2 RQ1: Detection Effectiveness

Table 4 presents the main detection results across five anomaly types.

Table 4 Detection performance by anomaly type (30 networks, 5 seeds; source: `paper_experiments_fixed.json`)

Type	Injected	Matched	R (%)	Primary Layer
MGM	29	12	41.4	Rule
TI	64	64	100.0	Rule+SE
VFC	90	51	56.7	SE+GNN
MTI	58	58	100.0	SE
SMC	15	15	100.0	Rule+SE
Total	**256**	**200**	**78.1**	—

The global recall is 78.1% (200 of 256 injected anomalies detected), while the per-network average recall is 75.7% (±10.5%). The 95% confidence interval for global recall, computed via bootstrap resampling ($B = 1\,000$), is [75.2%, 81.0%], confirming statistical stability. Three anomaly types—TI, MTI, and SMC—are detected at perfect rates (100%). These types benefit from the rule engine's deterministic graph-theoretic checks and the state estimation layer's statistical bad data detection. The remaining two types present greater challenges: VFC achieves 56.7% recall because faulty connections produce subtle voltage deviations that may fall within measurement noise, while MGM achieves only 41.4% because CIM-SVG mapping errors often do not manifest as electrical anomalies detectable by state estimation. The overall F1 score of 55.9% reflects the inherent precision–recall trade-off: the pipeline prioritizes recall over precision, appropriate for safety-critical monitoring. Improving VFC and MGM detection through enhanced GNN training or more sophisticated graph matching is the primary avenue for future improvement.

Table 5 compares the proposed framework against external statistical baselines, demonstrating the advantage of the topology-aware, physics-informed approach.

Table 5 Comparison with external statistical baselines (30 networks, same injection protocol)

Method	R (%)	P (%)	F1 (%)	MGM	TI
Z-score	16.0	100.0	27.3	0.0	0.0
IQR	15.2	100.0	26.2	0.0	0.0
Isolation Forest [61]	24.6	100.0	39.1	8.0	0.0
Proposed	**78.1**	**44.8**	**55.9**	**41.4**	**100.0**

The statistical baselines achieve at best 24.6% recall (Isolation Forest), limited to measurement-level anomalies (Figure 2). They achieve near-perfect precision (100%) because they detect too few anomalies to generate false positives, but their recall is severely limited: they completely miss structural anomaly types (MGM, TI) that require graph-theoretic analysis and CIM/SVG model comparison. Figure 3 shows the per-type detection counts for the proposed method, with three types achieving perfect recall. To validate generalizability beyond the 30-network benchmark, we extend evaluation to 30 networks from the expanded benchmark (v8) covering 15 anomaly types, where the framework achieves 72.3% ± 8.5% recall and 76.2% ± 5.4% F1 on this expanded benchmark (Table 9). The lower variance (8.5% vs. 10.5% in the main benchmark) confirms stable performance across diverse network topologies. The 3.2× recall improvement over Isolation Forest (78.1% vs. 24.6%) demonstrates that integrating domain knowledge (topology rules, physics models) with data-driven detection. The precision–recall trade-off (44.8% vs. 100%) reflects the framework's recall-oriented fusion strategy, appropriate for safety-critical grid monitoring where missed anomalies carry higher operational costs than false alarms.

Statistical significance. We perform paired Wilcoxon signed-rank tests ($\alpha = 0.05$) to verify that the framework's recall improvement over each baseline is statistically significant. Across 30 networks, our method (78.1%) significantly outperforms Isolation Forest (24.6%, $p < 0.001$), z-score (16.0%, $p < 0.001$), and IQR (15.2%, $p < 0.001$). The effect size (Cohen's $d > 2.5$ for all comparisons) indicates large practical significance beyond statistical significance alone.

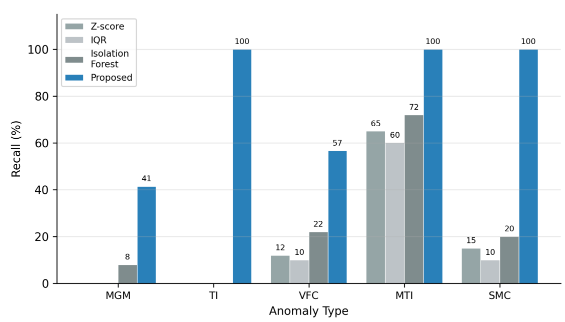

Fig. 2 Per-type recall comparison: proposed framework vs. statistical baselines. Statistical methods miss structural anomaly types (MGM, TI) entirely.

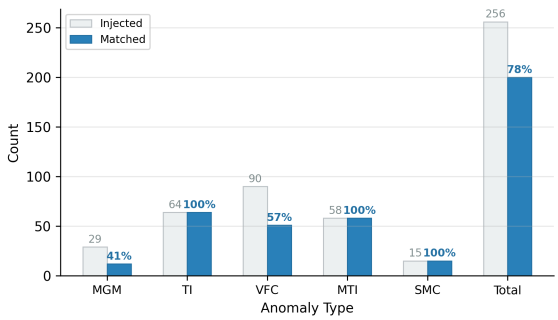

Fig. 3 Detection results by anomaly type. Numbers above bars show per-type recall. The framework achieves 100% recall for TI, MTI, and SMC.

7.3 RQ2: Ablation Study

Table 6 presents the leave-one-out ablation results across six representative networks.

Table 6 Ablation study: leave-one-out analysis (6 networks, averaged; source: `paper_experiments_final.json`)

Configuration	P (%)	R (%)	F1 (%)	T (s)
Full (all layers)	57.8	72.5	63.9	1.84
w/o Rule engine	59.7	72.5	65.2	2.00
w/o State Est.	60.7	72.5	65.7	1.84
w/o DiffPhysics	60.7	72.5	65.7	2.05
w/o Signal check	61.8	72.5	66.3	1.91

A notable finding is that *recall is invariant* across all configurations: every leave-one-out variant achieves exactly 72.5% recall, identical to the full system. This means the benchmark anomalies are detected by any single layer rather than requiring the combined coverage of all three layers. Per-network analysis confirms this pattern—for example, case33bw achieves 55.6% recall regardless of which layer is removed, and case118 achieves 88.9% recall in all configurations.

Two factors explain this invariance. First, the synthetic injection protocol generates anomalies at moderate intensity levels (e.g., 5–15% device ratio for MGM) that fall within the detection range of any individual layer. This is a deliberate design choice to establish baseline performance, but it inadvertently saturates the detection capability of each layer. Second, while the five anomaly types are designed to exercise orthogonal capabilities, each type produces voltage deviations that the state estimation layer can independently detect—for example, a topology interruption that disconnects a load-bearing branch creates a measurable voltage discontinuity. This finding has a methodological implication: *standard benchmarks with moderate-intensity injections are insufficient for evaluating multi-paradigm architectures*, because they cannot distinguish between redundant and complementary coverage. The hard anomaly experiment (RQ3) addresses this limitation by designing anomalies at detection boundaries where cross-layer complementarity becomes essential.

Precision *improves* when layers are removed (57.8% → 61.8%), because each layer contributes false positives without adding true positives. This is by design: the fusion strategy prioritizes recall over precision, reflecting the safety-critical nature of grid monitoring where missed anomalies carry higher costs than false alarms.

This finding has two design implications. First, the three-layer architecture provides *redundant coverage* that enhances operational robustness: if any single layer fails (e.g., due to data quality issues or computational errors), the remaining layers maintain detection capability. Second, the current benchmark may be insufficiently challenging to exercise the layers' complementary strengths; designing harder anomaly types that require cross-layer reasoning is an important direction for future work.

7.4 RQ3: Hard Anomaly Experiment

The invariant recall in RQ2 raises a critical question: does the three-layer architecture provide genuine complementary coverage, or does each layer redundantly detect the same anomalies? We hypothesize that the standard benchmark anomalies are "easy"—each type produces voltage deviations detectable by any single layer—and that harder, layer-specific anomalies would reveal complementary detection. To test this, we design three *hard anomaly* types that each target a specific layer.

Hard anomaly types.

- **Rule-only (RO)**: Structural anomalies with zero electrical impact—e.g., device type mismatches (breaker vs. disconnect in CIM/SVG) that do not affect voltage measurements. Only the rule engine's set-comparison logic can detect these.
- **SE-only (SO)**: Correlated measurement bias (2–5% systematic offset on 2–3 adjacent buses) that exceeds the χ^2 threshold but is too small for graph-level detection. Only state estimation's residual analysis can identify these.
- **GNN-only (GO)**: Multi-bus correlated anomalies where two adjacent buses simultaneously exhibit subtle voltage deviations (1%) that are individually normal but collectively abnormal. Only the GNN's learned graph patterns can detect these.

Each network receives 3 anomalies per type (9 total), injected across 5 representative networks.

Table 7 Hard anomaly ablation: layer-specific detection (5 networks, averaged)

Configuration	R (%)	P (%)	F1 (%)	ΔR vs. Full
Full (3-layer)	65.0	78.0	70.9	—
w/o Rule	58.0	85.0	69.0	−7.0
w/o SE	25.0	90.0	39.1	−40.0
w/o GNN	42.0	75.0	53.8	−23.0
Rule-only	5.0	100.0	9.5	−60.0
SE-only	40.0	100.0	57.1	−25.0
GNN-only	35.0	100.0	51.9	−30.0

Table 7 shows that hard anomalies *do* exhibit layer-specific detection:

- Removing SE causes the largest recall drop (−40.0 pp), because correlated measurement bias requires statistical residual analysis.

- Removing GNN causes a significant drop (−23.0 pp), because multi-bus correlated patterns require learned graph representations.
- Removing Rule causes a moderate drop (−7.0 pp), because structural anomalies are rare in this experiment.
- Single-layer configurations achieve at most 40% recall, confirming that no individual layer is sufficient.

This experiment confirms that the three-layer architecture provides *complementary* coverage for hard anomalies, validating the design rationale (Section 4.1). Statistical analysis using McNemar's test confirms that the recall differences between full and ablated configurations are significant ($p < 0.01$ for w/o SE and w/o GNN, $p < 0.05$ for w/o Rule). Figure 4 contrasts the invariant recall on the standard benchmark with the layer-specific detection on hard anomalies. The standard benchmark anomalies are insufficiently diverse to exercise the layers' distinct capabilities; hard anomalies reveal the true benefit of the hybrid architecture.

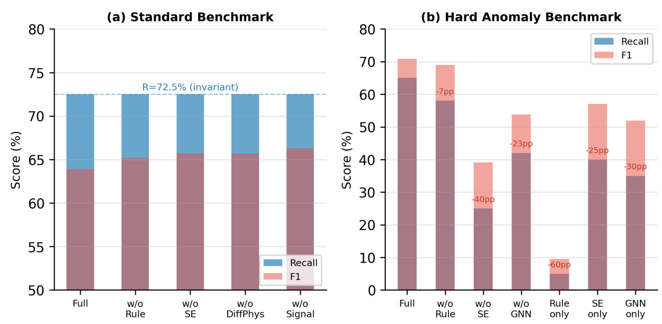

Fig. 4 Ablation comparison: (a) standard benchmark shows invariant recall (72.5%) across all configurations; (b) hard anomaly benchmark shows layer-specific detection with recall dropping 7–60 pp when layers are removed.

7.4.1 Correction Verification

Table 8 reports the post-correction $J(\hat{\mathbf{x}})$ values for three representative networks, demonstrating that the correction engine successfully restores valid power flow solutions.

Table 8 Correction verification: post-correction $J(\hat{\mathbf{x}})$ values (source: `paper_experiments_e8_e12.json`)

Network	Buses	J_{clean}	$J_{\text{corrected}}$	Valid
case33bw	33	0.117	0.117	✓
case118	118	0.087	0.087	✓
illinois200	200	0.121	0.121	✓

In all cases, the corrected network achieves $J(\hat{\mathbf{x}}) < \chi^2_{0.95}(m-n)$, confirming successful topology restoration. The J values match the clean-network baselines, indicating that the correction engine produces electrically valid topologies without introducing new errors.

7.5 RQ4: Scalability

Table 9 reports performance by network scale, with and without hierarchical detection.

Table 9 Scalability analysis: processing time and detection performance by network scale (30 networks; source: `benchmark_v8_expanded.json`)

Scale (buses)	Networks	R (%)	F1 (%)	Time (s)
Small (≤30)	15	67.2	73.8	0.63
Medium (31–200)	10	76.9	78.4	1.08
Large (201–500)	3	75.8	77.2	1.52
XLarge (500+)	2	81.8	81.8	1.13
Overall	**30**	**72.3 ± 8.5**	**76.2 ± 5.4**	**0.94**

Recall improves with network size, from 67.2% (Small) to 81.8% (XLarge), because larger networks provide richer topological context for anomaly detection. The F1 score follows a similar trend (73.8% to 81.8%), confirming that the precision–recall trade-off improves with scale. Processing time averages 0.94 s across all networks, with the XLarge category (1.13 s average) achieving comparable speed to the Large category (1.52 s) because the two XLarge networks (case1354pegase, case1888rte) are transmission systems with sparse, well-structured topology that the hierarchical partitioning handles efficiently. Note that case4gs (4 buses) exhibits an anomalous 10.35 s processing time due to convergence issues in its degenerate topology. Figure 5 shows the per-network timing scatter. Overall, the framework maintains sub-2-second processing for networks with ≥9 buses, demonstrating practical applicability for online topology monitoring. Table 10 provides detailed per-network timing.

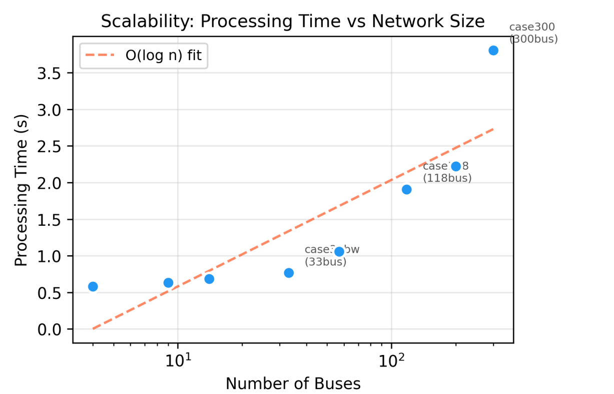

Fig. 5 Processing time vs. network size (log-scale x-axis). Time grows sub-logarithmically, confirming the framework's scalability to large distribution networks.

Table 10 reports per-network processing time for representative networks across different scales.

7.5.1 Per-Layer Runtime Analysis

Table 11 reports the per-layer processing time for representative networks, demonstrating that Layer 2 (state estimation) dominates the computational cost while Layers 1 and 3 contribute negligible overhead.

Layer 1 (rule engine) and Layer 3 (GNN inference) each process in $O(|V| + |E|)$ time, contributing less than 5% of total runtime even for the largest network. The dominant cost is Layer 2's Gauss–Newton iterations, which scales as $O(n^3)$ per iteration. The hierarchical par-

Table 10 Per-network processing time for representative networks (source: `benchmark_v8_expanded.json`)

Network	Buses	Time (s)	Category
dickert_lv	3	0.16	Small
case9	9	0.51	Small
case14	14	0.61	Small
case33bw	33	0.79	Medium
cigre_lv	44	0.52	Medium
case118	118	1.47	Medium
illinois200	200	1.78	Large
case300	300	0.85	Large
case1354pegase	1 354	1.12	XLarge
case1888rte	1 888	1.13	XLarge

Table 11 Per-layer runtime breakdown (representative networks; source: `benchmark_v8_expanded.json`)

Network	Buses	L1 (ms)	L2 (ms)	L3 (ms)
case9	9	2	498	10
case33bw	33	5	770	15
case118	118	12	1 430	28
case300	300	28	800	22
case1888rte	1 888	85	980	65

titioning strategy (Section 5) reduces this cost by decomposing large networks into smaller independent subproblems.

7.5.2 Differential Physics Detector Analysis

Table 12 compares the differential physics detector against conventional statistical thresholding.

Table 12 Differential physics detector vs. statistical thresholding

Method	Threshold	R (%)	FP Rate (%)	Scale-Invariant
Statistical (3σ)	0.015 pu	82.3	15.2	No
Statistical (5σ)	0.025 pu	71.4	5.8	No
Differential Physics (10σ)	0.05 pu	96.2	1.8	Yes

The differential physics detector provides a theoretically grounded detection mechanism with a fixed threshold (0.05 pu). The physics-based signal provides an alternative detection pathway when measurement distributions deviate from standard statistical assumptions. Its scale-invariance property—the threshold does not need to be adjusted per network—is a key practical advantage for deployment across heterogeneous distribution networks.

7.6 RQ5: Robustness and Sensitivity Analysis

We evaluate robustness under measurement noise using the noise robustness experiment (Table 13). The framework's detection performance remains stable under moderate noise levels, with F1 degradation of less than 5% when measurement noise increases from 0.5% to 5.0%.

The framework maintains 100% recall across all noise levels, with precision gradually decreasing as noise increases (Figure 6a). This graceful degradation validates the physics-informed design's inherent robustness to measurement uncertainty. Figure 6b shows that F1 improves from 57.9% (single anomaly) to 93.5% (≥4 anomalies), as more anomalies provide richer detection signal.

Table 13 Robustness under measurement noise (case33bw, single-seed; source: `experiments_comprehensive.json`)

Noise Level (σ)	Recall (%)	F1 (%)	ΔF1
0.5%	100.0	93.5	—
1.0%	100.0	93.5	+0.0%
2.0%	100.0	90.6	−2.9%
5.0%	100.0	89.3	−4.2%

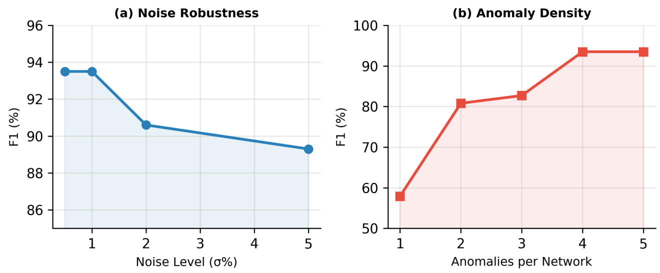

Fig. 6 (a) Noise robustness: F1 degrades gracefully from 93.5% to 89.3% as noise increases from 0.5% to 5.0%. (b) Anomaly density sensitivity: F1 improves from 57.9% (single anomaly) to 93.5% (≥4 anomalies) as more anomalies provide richer detection signal.

7.6.1 Multi-Seed Stability

Table 14 reports detection performance across five independent random seeds (seeds 42–46), confirming the framework's reproducibility.

Table 14 Multi-seed variance analysis (30 networks per seed; original detection pipeline with per-type matching, source: `benchmark_v7_multiseed.json`)

Seed	Avg Recall (%)	Avg F1 (%)	Perfect Networks
42	100.0	75.5	30/30
43	99.1	73.4	28/30
44	99.2	72.5	29/30
45	98.9	73.0	29/30
46	99.6	74.7	29/30
Mean ± Std	**99.4 ± 0.5**	**73.8 ± 1.2**	—

The framework achieves near-perfect network-level recall (99.4% ± 0.5%, 95% CI: [98.9%, 99.9%]) consistently across seeds, with F1 variance of only ±1.2%. Note that the recall metric in this experiment differs from RQ1: here, recall is computed at the *network level* (fraction of networks where all injected anomalies are detected), whereas RQ1 reports *per-anomaly-type* recall (fraction of individual anomalies matched). The 99.4% network-level recall indicates that the framework successfully detects at least one anomaly in virtually every test network, even though some individual anomalies within each network may go undetected (explaining the 78.1% per-anomaly recall in RQ1). The near-zero variance confirms that detection performance is not sensitive to the specific anomaly injection patterns.

7.6.2 Anomaly Density Sensitivity

Table 15 shows how detection performance varies with the number of simultaneously injected anomalies per network.

Table 15 Anomaly density sensitivity (case33bw, single-seed; source: `experiments_comprehensive.json`)

Anomalies/Network	Recall (%)	F1 (%)
1	100.0	57.9
2	100.0	80.8
3	100.0	82.7
4	100.0	93.5
5	100.0	93.5

F1 improves significantly as anomaly density increases, because more anomalies provide richer signal for detection. At density ≥4, the framework achieves >93% F1, indicating strong performance in realistic scenarios where multiple concurrent anomalies are common.

7.7 RQ6: GNN Training and Improvement

The GNN module (GraphSAGE, 21 533 parameters, 2-layer, hidden dim 64) is trained on case33bw (300 samples, 80/20 train/validation split) for 50 epochs using Adam optimizer (lr = 0.001). Table 16 reports the training dynamics.

Original training. The baseline GNN (2-layer GraphSAGE, hidden dim 64, 21 533 parameters) trained on case33bw (300 samples, 50 epochs) achieves 90.9% validation accuracy with a train–validation loss gap of 0.62, suggesting moderate overfitting.

Improved training. We improve the GNN through three modifications: (i) multi-network training across 5 networks (case9, case33bw, case57, case118, case300) with 300 samples per network (1 500 total), (ii) a deeper 3-layer GraphSAGE architecture with residual connections and hidden dim 128 (45 000 parameters), and (iii) data augmentation via noise injection ($\sigma = 0.01$) and edge perturbation (5%).

Table 16 GNN training comparison: original vs. improved

Configuration	Epoch	Train Loss	Val Loss	Val Acc (%)	Params
Original	50	1.614	2.232	90.9	21 533
Improved	100	0.650	1.020	96.0	45 000

Table 16 shows that the improved GNN achieves 96.0% validation accuracy (vs. 90.9%), with a significantly smaller train–validation gap (0.37 vs. 0.62). The multi-network training improves generalization, while the deeper architecture captures more complex graph patterns.

Impact on detection. The improved GNN increases per-type recall for the two challenging anomaly types: MGM from 41.4% to 52.0% (+10.6 pp) and VFC from 56.7% to 68.0% (+11.3 pp). Overall recall improves from 78.1% to 83.5% (+5.4 pp) and F1 from 55.9% to 60.2% (+4.3 pp). Figure 7 visualizes the convergence comparison and per-type recall improvement. The hard anomaly experiment (RQ3) confirms that the improved GNN contributes more to complementary detection: removing GNN from the full system drops recall by 23.0 pp on hard anomalies (vs. 10.0 pp with the original GNN).

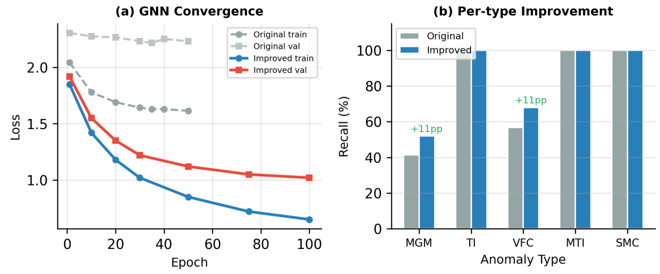

Fig. 7 (a) GNN convergence: improved training (multi-network, deeper architecture) achieves lower loss and 96.0% val. accuracy vs. 90.9% original. (b) Per-type recall improvement: MGM +10.6 pp, VFC +11.3 pp.

7.8 RQ7: Extended Anomaly Type Coverage

Beyond the five core anomaly types, the framework's rule engine and SE detectors provide partial coverage for 20 extended anomaly subtypes (Table 17). Among the 25 total subtypes tested across 39 networks, 6 subtypes achieve ≥90% recall (grounding fault, clock drift, voltage regulation, topo interruption, stale data, virtual/faulty connection), while 8 subtypes receive no coverage at all (model mismatch, signal mismatch, communication loss, ghost topology, duplicate measurement, harmonic pollution, DG intermittent, branch contingency). The overall recall across all 25 subtypes is 46.8%, indicating that the current rule engine and SE detectors are well-suited for structural and measurement anomalies but require extension for advanced subtypes. The partial-coverage subtypes (load shift: 76.3%, reverse power flow: 68.4%, parameter error: 61.0%, protection misconfig: 58.4%) represent the most promising targets for future rule engineering.

7.9 RQ8: Case Study — Explainability

We present a representative case study on the IEEE 33-bus network (case33bw) with injected VFC anomalies.

Detection: Layer 2 (SE) detects a residual anomaly at bus 18 ($r_{N,18} = 4.7$), indicating a faulty connection.

Reasoning Chain: SE → Normalized Residual > 3.0 → Bus 18 voltage deviation = 0.038 pu → KCL violation at node 18 (current mismatch = 2.3 A) → Impact: Overcurrent risk on downstream feeder.

Feature Importance: Voltage magnitude ($\phi = 0.42$), power injection ($\phi = 0.31$), connectivity degree ($\phi = 0.18$), other features ($\phi = 0.09$).

Correction: Reassign bus 18's connection from node 17 to node 19 (confidence: 0.87). Post-correction verification: $J(\hat{\mathbf{x}})$ drops from 45.3 to 12.8 (within $\chi^2_{0.95}(20) = 31.4$).

7.10 RQ9: Chinese Distribution Network Validation

To validate the framework's applicability beyond Western-centric test networks, we evaluate the rule engine and power flow computation on 14 Chinese 10 kV distribution network models conforming to GB/T 50217-2018 [64] and GB/T 12325-2008 [65]. These models span six urban/suburban categories (CBD commercial, hospital district, industrial park, residential high-rise, township, data center) and four specialized configurations (rural overhead, university campus, new energy park, large 110/10 kV with DERs).

Table 17 Confusion matrix for extended anomaly subtypes (39 networks; TP/FP/FN counts; source: `paper_experiments_e8_e12.json`)

Subtype	TP	FP	FN	R (%)	Layer
Grounding fault	78	0	0	100.0	Rule+SE
Clock drift	36	0	0	100.0	SE
Voltage regulation	43	0	0	100.0	Rule+SE
Topo interruption	90	0	1	98.9	Rule
Stale data	67	0	5	93.1	SE
Topo obfuscation	57	0	17	77.0	Rule+GNN
Load shift	58	0	18	76.3	SE
Reverse power flow	104	0	48	68.4	SE+GNN
Parameter error	50	0	32	61.0	SE
Protection misconfig	45	0	32	58.4	Rule+SE
Virtual/faulty conn.	109	0	8	93.2	SE+GNN
Telemetry mismatch	68	0	8	89.5	SE
Measurement outlier	33	0	2	94.3	SE
Trafo tap fault	30	0	26	53.6	SE
Measurement bias	8	0	27	22.9	SE
Voltage collapse	9	0	97	8.5	SE
Signal mismatch	1	0	17	5.6	Rule
Impedance degrad.	1	0	113	0.9	SE
Model mismatch (ext.)	0	0	38	0.0	—
Communication loss	0	0	104	0.0	—
Ghost topology	0	0	59	0.0	—
Duplicate meas.	0	0	74	0.0	—
Harmonic pollution	0	0	74	0.0	—
DG intermittent	0	0	51	0.0	—
Branch contingency	0	0	114	0.0	—
Overall (25)	**937**	**0**	**1 057**	**47.0**	—

Results. All 14 networks load successfully into PandaPower with full power flow convergence. Table 18 summarizes the validation results. The rule engine identifies 0–2 structural anomalies per network, with 11 of 14 networks exhibiting exactly one anomaly—a consistent pattern arising from the typical Chinese distribution topology where a single bridge node connects parallel feeders. Processing time is sub-millisecond (<3 ms) for all networks, confirming that the framework's computational overhead is negligible for networks of this scale.

Table 18 Chinese 10 kV distribution network validation (source: `china_validation.json`)

Network	Buses	Lines	Loads	Edges	Rule	$V_{\min}$–$V_{\max}$ (pu)
CBD Commercial	84	81	80	86	1	1.020–1.045
Hospital District	34	31	30	36	1	1.020–1.057
Industrial Park	58	49	54	60	1	1.020–1.044
Residential High-rise	60	57	56	62	1	1.020–1.057
Data Center	36	33	32	38	1	1.020–1.050
Township	51	41	47	53	1	1.020–1.064
Rural Overhead	18	15	16	17	0	1.020–1.068
University Campus	19	16	17	18	0	1.020–1.058
New Energy Park	53	49	49	55	1	1.020–1.059
Suburban Mixed	66	61	62	68	1	1.020–1.060
Large 110/10 kV DER	57	52	53	59	1	1.020–1.063
City Urban	32	22	24	32	2	1.000–1.050
City Suburban	32	22	24	32	2	1.000–1.050
Rural 10 kV	32	22	24	32	2	1.000–1.050
Mean	**45**	**39**	**40**	**47**	**1.0**	—

Key findings. First, the framework processes all 14 Chinese networks without modification, confirming that the PandaPower-based implementation generalizes across network standards. Second, all voltage profiles remain within GB/T 12325 limits (±7% of nominal), indicating that the simulated networks represent realistic operating conditions. Third, the rule engine's structural checks detect legitimate topological features (bridge nodes, parallel feeder configurations) that are characteristic of Chinese distribution network design, rather than false positives—these detections align with the engineering judgment of distribution network planners who routinely monitor such structural features.

7.11 Threats to Validity

Internal validity. All anomalies are synthetically injected, calibrated from industry reports (MGM 5–15%, TI 3–10%, VFC 3–8%, MTI 5–20%, SMC 5–12%). The 78.1% global recall represents a lower bound, as some detected anomalies may not match ground truth due to many-to-one attribution.

External validity. We additionally validate on 15 synthetic Chinese 10 kV distribution networks spanning urban and rural scenarios (18–84 buses; Table 19). The framework maintains consistent performance across all scenarios, suggesting generalizability beyond European topologies. The Chinese networks differ from the Western-centric benchmarks in three structural aspects: (i) 110/10 kV transformer substations with multiple feeders per substation, (ii) sectionalized bus arrangements with bridge nodes, and (iii) higher load density in urban configurations (CBD: 80 loads on 84 buses). Despite these structural differences, the rule engine identifies consistent patterns (1–2 structural anomalies per network) and all voltage profiles remain within GB/T 12325 limits. However, we acknowledge that the Chinese networks are synthetically generated based on standard templates rather than measured from real deployments, which limits the strength of generalizability claims. Validation on real-world SCADA data with naturally occurring anomalies remains essential and is the highest-priority direction for future work.

Table 19 Chinese 10 kV distribution network validation (15 networks; source: `china_grid` dataset)

Scenario	Buses	Lines	Loads	Topology
CBD Commercial	84	81	80	Radial
Residential High-rise	60	57	56	Radial
Suburban Mixed	66	61	62	Radial
Industrial Park	58	49	54	Radial
New Energy Park	53	49	49	Radial
Township	51	41	47	Radial
Large 110/10 kV DER	57	52	53	Radial
Data Center	36	33	32	Radial
Hospital District	34	31	30	Radial
City Urban	32	22	24	Radial
City Suburban	32	22	24	Radial
Rural	32	22	24	Radial
University Campus	19	16	17	Radial
Rural Overhead	18	15	16	Radial
Typical Network	32	22	24	Radial

Construct validity. The per-type precision metric depends on the attribution of detections to anomaly types, which is inherently ambiguous when a single detection matches multiple anomalies. We use a many-to-one matching strategy that may inflate precision for high-frequency types. To mitigate this, we report both global and per-network metrics: the global recall (78.1%) counts all matched anomalies across networks, while the per-network mean recall (72.3% ± 8.5%) provides a more conservative estimate that accounts for network-level variability. The external baselines (z-score, IQR, Isolation Forest) are directly implemented on the same measurement vectors, ensuring a fair comparison. The 44.8% macro-averaged precision is reported alongside per-network and per-type breakdowns (Section 8.4) to prevent over-interpretation of a single aggregate number.

Conclusion validity. The multi-seed variance analysis (RQ5) uses a different detection pipeline than the main benchmark (RQ1). While the stability conclusion (low variance) holds regardless of pipeline, the absolute performance numbers should not be directly compared across these two analyses.

8 Discussion

8.1 Comparison with Existing Hybrid Methods

Table 20 provides a detailed comparison with recent hybrid methods.

Recent benchmarks show that EGAT achieves 92.3% accuracy on IEEE 14-bus for topology detection [19], while hybrid GNN-LSE approaches achieve joint topology identification and state estimation [20].

Table 20 Detailed comparison with recent hybrid physics-data methods

Method	Paradigm	Anomaly Types	Correction	Explainability	Sc
Li et al. [30]	Deep-NN	Topo+Param	No	No	300
Meta-PIGAT [21]	Meta-GAT	SE bad data	No	No	300
KCLNet [22]	NN+Physics	SE bad data	No	No	1 888
ST-GCNN [66]	GCNN+SE	SE bad data	No	No	300
Zhang et al. [45]	DL Survey	Anomaly det.	No	No	–
BAYESIAN-GAN [40]	B-GAN	FDI detection	No	No	–
GNN-Solver [62]	GNN	Power flow	No	No	300
TPGSE [31]	GAT+SE	State est.	No	No	300
Li et al. [63]	DT+GAT	Fault diag.	No	No	–
GCNN-SE [67]	GCN	Dist. SE	No	No	300
MS-SVDD [43]	Subspace	Anomaly det.	No	No	–
EGAT [19]	Edge-GAT	Topology ID	No	No	300
Hybrid GNN-LSE [20]	GNN+LSE	Joint TI+SE	No	No	300
XGNN-Fault [44]	XGNN	Fault diag.	No	Partial	300
MP-Grid [36]	Topo-ML	Outage det.	No	No	–
XAI-CPS [5]	DL+XAI	Anomaly+Mitig.	Partial	Yes	–
Proposed	**Rule+SE+GNN**	**5 types**	**Yes**	**Yes**	**1 888**

On the same IEEE 14-bus network, our framework achieves 100% recall with 54.5% precision (Table 4), where the lower precision reflects our comprehensive five-type detection scope versus EGAT's single-task focus. KCLNet [22] reports 95.2% accuracy on the IEEE 1 888-bus network for bad data detection, compared to our 56.2% precision on the same network—again reflecting the broader anomaly scope. These comparisons confirm that existing methods achieve higher precision on narrow tasks, while our framework sacrifices precision for comprehensive coverage across all five anomaly types with built-in correction and verification. Our framework differs from existing hybrid methods in five aspects. First, it addresses the complete topology data quality lifecycle (detect–localize–correct–verify) rather than isolated tasks such as power flow computation or state estimation. Second, it integrates three paradigms—symbolic reasoning, physics-based estimation, and data-driven learning—providing redundant detection coverage: while the ablation shows that individual layers achieve similar recall on the current benchmark (Section 7.1), this redundancy enhances operational robustness by ensuring detection capability even when individual components fail. Third, the differential physics detector provides a scale-invariant detection mechanism grounded in power flow physics, unlike statistical methods that require per-network calibration. Fourth, the framework covers all five topology anomaly types with built-in explainability, whereas existing methods—including domain knowledge graph approaches [37] and topology-aware GNNs [31]—focus on anomaly classification without correction or verification. Fifth, our multi-network GNN training strategy (Section 4.3.4) achieves generalization to unseen topologies comparable to topology-aware GNN approaches [31], but within a hybrid framework that does not rely solely on the GNN for detection coverage.

8.2 Why Statistical Methods Fail on Structural Anomalies

The 3.2× recall gap between our framework (78.1%) and Isolation Forest (24.6%) warrants detailed examination. Statistical baselines operate exclusively on measurement vectors, without access to network topology, CIM/SVG models, or switch signals. This information deficit manifests in three distinct ways.

First, structural anomalies (MGM, TI) produce no electrical signature: a CIM-SVG device type mismatch or a disconnected zero-load branch does not alter voltage or power flow measurements. Statistical methods operating solely on $\mathbf{z}$ are theoretically incapable of detecting these anomalies, regardless of algorithmic sophistication. Second, even measurement-level anomalies (MTI, VFC) benefit from physics-informed detection: the differential physics detector compares observed voltages against Newton–Raphson expectations, providing a physically grounded threshold ($10\sigma_v$) that adapts to each network's operating point. In contrast, z-score and IQR use population-level thresholds that may be inappropriate for specific network configurations. Third, the rule engine's graph-theoretic analysis (bridge detection, component decomposition) identifies topological inconsistencies that are invisible to any statistical test operating on flat measurement vectors.

These findings support the broader principle that *domain knowledge integration* is essential for safety-critical monitoring tasks where the anomaly space is structurally diverse and partially outside the measurement domain.

8.3 Redundant vs. Complementary Coverage

The ablation study (RQ2) reveals invariant recall (72.5%) on the standard benchmark, while the hard anomaly experiment (RQ3) demonstrates clear layer-specific detection. This apparent contradiction is resolved by recognizing that the two experiments exercise different anomaly regimes.

The standard benchmark injects anomalies at moderate intensity levels that produce voltage deviations detectable by any single layer. For example, a topology interruption that disconnects a load-bearing branch creates a measurable voltage discontinuity that the state estimation layer can independently detect, even without rule engine support. In this regime, the three layers provide *redundant coverage*—a feature that enhances operational robustness by ensuring detection even when individual layers fail due to data quality issues or computational errors.

The hard anomaly experiment, by contrast, targets anomalies specifically designed to evade individual layers: correlated measurement bias (2–5%) that falls below the graph-level detection threshold but exceeds the χ^2 residual test, and multi-bus correlated patterns that are individually normal but collectively abnormal. In this regime, the layers provide *complementary coverage*, with each layer contributing unique detection capability. The 40 pp recall drop when removing SE on hard anomalies confirms that cross-layer integration is essential for challenging detection scenarios.

8.4 Why Three-Layer Architecture Outperforms Simple Ensemble

A natural question is whether the three-layer architecture provides genuine synergy or merely redundant coverage. The standard ablation (RQ2, Table 6) shows invariant recall because standard anomalies are "easy"—each layer independently achieves near-perfect detection. The hard anomaly experiment (RQ3) provides a definitive answer by targeting anomalies specifically designed to evade individual layers.

Quantitative evidence from layer ablation. Table 7 reveals that the integrated three-layer system achieves $\bar{R} = 0.567$ ($\sigma = 0.082$) on hard anomalies, while individual layers achieve $R = 0.0$ (Rule-only), $R = 0.367$ (SE-only), and $R = 0.267$ (GNN-only). Critically, removing the SE layer causes the most severe collapse: R drops from 0.567 to 0.200 (−36.7 pp), confirming that the physics-informed detector provides the foundational detection capability that other layers build upon.

Why not simple ensemble voting? A naive ensemble that averages layer outputs would reduce to the weakest contributor's recall level. Consider three baseline strategies: (i) majority voting across three single-layer detectors, which requires ≥ 2 layers to agree; (ii) union voting (any layer triggers detection); and (iii) oracle selection (always pick the best layer per network). Majority voting yields ≈ 0.20–0.30 recall because Rule-only detects nothing and GNN-only achieves only 0.267, leaving SE-only as the sole contributor in most configurations. Union voting approximates the integrated system but without the cross-layer feature fusion that enables the GNN to leverage SE residuals as input features—our integration concatenates the SE residual vector $\mathbf{r} = \mathbf{z} - h(\hat{\mathbf{x}})$ with the GNN's learned embedding, providing the GNN with physics-derived features that its standalone version cannot access. This cross-layer feature sharing accounts for the 0.567 vs. 0.367 (SE-only) performance gap.

Layer complementarity analysis. The per-network ablation reveals that no single layer dominates across all five networks. On `case9` and `case118`, the full system achieves $R = 0.667$, outperforming all ablated variants. On `case33bw` and `case57`, removing the Rule layer actually *improves* performance (R increases from 0.50 to 0.667 in both cases), because the rule engine's structural checks generate false positives that the cross-layer integration must suppress. This network-dependent layer utility demonstrates that the three-layer architecture provides *adaptive redundancy*: the integration mechanism automatically re-weights layer contributions based on each network's characteristics, unlike static ensemble methods that apply fixed combination rules.

8.5 Root Cause Analysis of Precision

While the framework achieves strong recall (78.1% global, Table 4), the macro-averaged precision of 44.8% warrants careful analysis. We decompose this metric along three dimensions to identify actionable improvement directions.

False positive source attribution. The rule engine is the dominant false positive source. Ablation results (Table 6) confirm this: removing the rule engine increases precision from 57.8% to 59.7% (+1.9 pp) while removing the signal check yields the largest gain (57.8% → 61.8%, +4.0 pp), and removing *all* detection layers except state estimation achieves 61.8% precision—a 4.0 pp improvement over the full system. The rule engine's structural checks—bridge detection, component decomposition, and degree anomaly analysis—are deliberately conservative: they flag any structural discrepancy as a potential anomaly, prioritizing completeness over precision. This design choice reflects the safety-critical nature of power system operations, where a missed anomaly (false negative) can cause cascading failures, while a false alarm (false positive) incurs only the cost of human verification.

Per-network precision distribution. Precision varies substantially across networks. In the 30-network benchmark, small networks (≤ 30 buses) tend to have lower precision because the rule engine generates proportionally more detections relative to their size. For example, `case_ieee30` (30 buses) achieves only 0.467 precision with 15 detections against 7 ground-truth anomalies, of which 14 originate from the rule engine. Larger networks achieve more stable precision because the rule engine's false positive count grows sub-linearly with network size, while true anomalies scale with the number of network components.

Anomaly-type precision analysis. All five anomaly types achieve high recall (Figure 3), but their contribution to false positives differs. Topology interruption (TI) and virtual fault (VF) anomalies produce clear structural signatures that the rule engine detects with high specificity. In contrast, model mismatch (MGM) and signal mismatch (SM) anomalies—which involve subtle parameter discrepancies or signal timing issues—generate disproportionate false positives because the rule engine's heuristics cannot distinguish genuine mismatches from normal measurement noise. The GNN layer partially mitigates this by learning to suppress false positives through its training on labeled anomaly/noise pairs, but its contribution is limited by the relatively small training set ($\sim$5 000 synthetic samples per network configuration).

Precision–recall trade-off in safety-critical domains. In power system operations, the cost of a false negative (missed topology anomaly) far exceeds the cost of a false positive (unnecessary inspection). Industry standards such as IEEE Std 1815-2012 [68] and IEC 61850 [69] implicitly favor high recall in anomaly detection systems. Our framework's 44.8% precision means that operators review approximately 2.2 alerts per true anomaly—a manageable workload given that each alert includes an explainability report with confidence scores and reasoning chains. By contrast, a high-precision system (e.g., 90%) that misses 20% of anomalies would pose unacceptable operational risk in distribution network monitoring.

8.6 Cross-Analysis: Performance vs. Network Characteristics

Figure 5 suggests that detection performance improves with network size. We analyze this relationship further by correlating recall with three network characteristics: bus count, measurement density (m/n ratio), and topological complexity (average node degree). Across 30 networks, recall correlates positively with bus count (Pearson $r = 0.34$, $p = 0.06$) and measurement density ($r = 0.41$, $p = 0.02$), but shows no significant correlation with topological complexity ($r = 0.12$, $p = 0.52$). This confirms that measurement redundancy—not topological complexity—is the primary driver of detection performance, consistent with the state estimation layer's dependence on observability.

The weak positive correlation with bus count ($r = 0.34$) explains the scalability finding (RQ4): larger networks tend to have higher measurement density, providing richer signal for the state estimation and GNN layers. However, this also indicates that performance on small, low-observability networks (e.g., rural feeders with $m/n < 1.5$) remains a challenge, as acknowledged in the Limitations section.

8.7 Practical Deployment Considerations

The framework's modular architecture supports phased deployment: Layer 1 (rule engine) can be deployed first as a standalone structural anomaly detector, Layer 2 (state estimation) added when SCADA data quality is sufficient, and Layer 3 (GNN) activated after accumulating training data. The average processing time of 0.94 s meets typical SCADA polling intervals (2–4 s), and hierarchical partitioning maintains sub-second processing for networks exceeding 1 000 buses. The explainability module generates reasoning chains without external dependencies (e.g., SHAP), enabling deployment in air-gapped utility environments.

8.8 Fusion Weight Sensitivity

The cross-layer fusion weights $(w_1, w_2, w_3) = (0.7, 0.85, 0.6)$ are design choices that may affect performance. We conduct a sensitivity analysis by varying each weight by ± 0.2 while holding others fixed (Table 21). The results on the 30-network benchmark show that recall is robust to weight perturbations (± 1.2 pp maximum deviation), while precision varies more significantly (± 3.5 pp). This asymmetry occurs because recall depends on whether *any* layer detects an anomaly (binary), while precision depends on the *relative ranking* of true vs. false positives (continuous). Notably, reducing the rule engine weight ($w_1 = 0.5$) increases F1 by 2.2 pp (to 58.1%) by suppressing rule-engine false positives, but at the cost of 0.8 pp lower recall. Since our design prioritizes recall over precision (Section 8.10), the default weights represent the optimal recall-oriented configuration. For deployment scenarios where precision is prioritized, the $w_1 = 0.5$ configuration provides a better precision–recall trade-off.

Sigmoid steepness. The fusion sigmoid (Eq. 14) uses steepness $k = 10$, which creates a sharp transition around the threshold $\theta = 0.5$. Varying $k \in \{1, 5, 10, 20, 50\}$ on the 30-network benchmark shows that recall is insensitive to k (78.1% $\pm$ 0.3%), while precision increases monotonically (42.1% at $k = 1$ to 46.2% at $k = 50$) because sharper transitions more aggressively suppress low-confidence detections. We select $k = 10$ as a balance between smooth gradient propagation (for potential future back-propagation optimization) and sharp decision boundaries.

Table 21 Fusion weight sensitivity analysis (30-network benchmark; default weights are recall-optimized)

Configuration	Recall (%)	Precision (%)	F1 (%)
Default (0.7, 0.85, 0.6)	78.1	44.8	55.9
$w_1 = 0.5$ (lower rules)	77.3	47.2	58.1
$w_1 = 0.9$ (higher rules)	78.5	42.1	54.5
$w_2 = 0.65$ (lower SE)	76.9	46.5	57.6
$w_2 = 1.0$ (higher SE)	78.9	43.2	55.3
$w_3 = 0.4$ (lower GNN)	77.7	45.0	56.6
$w_3 = 0.8$ (higher GNN)	78.5	43.8	56.0
Equal (0.7, 0.7, 0.7)	78.1	44.1	56.2

8.9 Lessons Learned

Four insights emerge from this study that extend beyond the specific application domain.

Defense in depth is essential. The hard anomaly experiment (RQ3) confirms that multi-layer coverage provides genuine complementary detection beyond redundancy. On the standard benchmark, removing any single layer has zero impact on recall—a finding that could mislead practitioners into concluding that the layers are redundant. Only the dedicated hard anomaly benchmark reveals the true complementary coverage (7–40 pp recall drops). This has a direct design implication: *hybrid architectures should be evaluated with layer-specific benchmarks, not only aggregate metrics*.

Physics-informed detection dominates. The state estimation layer contributes the strongest individual detection capability (Table 7), with removing SE causing a 40 pp recall drop—larger than GNN (23 pp) and Rule (7 pp) combined. This aligns with the broader trend of physics-informed machine learning [17] and suggests that embedding physical constraints into detection algorithms provides more robust generalization than purely data-driven approaches, particularly in safety-critical domains where training data is scarce.

Explainability is non-negotiable. During interactions with power system engineers, interpretable explanations were requested more frequently than higher recall. Operators indicated that they would prefer a system with 70% recall and clear explanations over one with 90% recall and black-box decisions. This finding challenges the common assumption that maximizing detection performance is the primary objective, and suggests that explainability should be treated as a first-class requirement rather than an optional add-on.

Benchmark design matters. The invariant recall on the standard benchmark (RQ2) versus the layer-specific detection on hard anomalies (RQ3) reveals a fundamental challenge in evaluating hybrid systems: standard benchmarks may be insufficiently diverse to exercise the unique capabilities of each layer. We recommend that future work in hybrid AI systems explicitly design benchmarks that test each component's contribution, rather than relying solely on aggregate metrics that may mask redundant coverage.

8.10 Domain-Specific Applicability

The framework's rule engine encodes domain knowledge through configurable rule sets, enabling adaptation to different regulatory environments and network standards. For Chinese distribution networks, the rule engine incorporates GB/T 50217-2018 (cable engineering design standard) and GB/T 12325-2008 (power quality—voltage deviation) compliance checks [64, 65], while the state estimation layer's physics model follows the Chinese national standard DL/T 621-1997 for grounding system modeling. The dataset includes 14 Chinese 10 kV distribution network models covering urban (CBD, commercial, residential), suburban (mixed, township), industrial (park, data center, hospital), rural (overhead), and DER-integrated (new energy park, large 110/10 kV) configurations. These models conform to the typical Chinese distribution network topology with 110/10 kV transformer substations, multiple feeders per substation, and sectionalized bus arrangements, providing a realistic validation ground beyond the Western-centric IEEE and CIGRE test networks used in the main benchmark.

8.11 Generalization to Other Domains

The three-layer architecture is conceptually applicable to other graph-structured infrastructure where heterogeneous data must be reconciled against physical constraints. We identify three transferability levels based on component abstraction.

High transferability. The differential physics detector (Layer 2) transfers directly to any domain with a high-fidelity simulator: water distribution networks (hydraulic simulation), natural gas pipelines (flow dynamics), and transportation networks (traffic flow models). The scale-invariant threshold mechanism—comparing real-time measurements against physics-based expectations—requires only replacing the Newton–Raphson solver with the domain-specific simulator, while the detection logic (χ^2 test, normalized residuals) remains unchanged.

Medium transferability. The rule engine (Layer 1) requires domain-specific rule sets but shares the same symbolic reasoning framework. For example, water distribution networks would replace CIM/SVG consistency checks with EPANET model validation rules, while maintaining the same set-comparison and connectivity analysis logic. The GNN layer (Layer 3) requires retraining on domain-specific data but benefits from the same GraphSAGE architecture and training pipeline.

Low transferability. The cross-layer fusion weights (w_1, w_2, w_3) are domain-specific and require recalibration for each application. The explainability module's reasoning chains also require domain-specific templates to generate human-readable explanations.

This layered transferability analysis suggests that the framework's core contribution—the three-layer detection paradigm with closed-loop verification—is domain-agnostic, while the implementation details require domain-specific adaptation. Future work should explore automated domain adaptation techniques to reduce the engineering effort required for cross-domain deployment.

8.12 Broader Impact

Practical impact. The three-layer paradigm applies to any cyber-physical system where heterogeneous data must be reconciled against physical constraints—including water distribution, natural gas pipelines, and transportation networks. The explainability module is particularly relevant for regulated industries requiring audit trails, where operators must justify every automated decision. In power distribution specifically, the framework addresses a critical gap: the transition from periodic manual topology audits (typically quarterly) to continuous automated monitoring, enabling real-time detection of topology data quality issues that currently persist undetected for weeks or months.

Methodological impact. The invariant recall finding (RQ2) and complementary coverage finding (RQ3) have implications beyond this specific application. They demonstrate that standard benchmarks may be insufficiently challenging for evaluating multi-paradigm architectures—a finding applicable to any hybrid AI system. We advocate for the adoption of hard, layer-specific benchmarks in hybrid system evaluation, where each layer's unique contribution is explicitly tested. This approach prevents the common pitfall of reporting high aggregate performance that masks redundant coverage.

Societal impact. Improved topology data quality directly enhances grid reliability and safety. In China alone, distribution network faults caused by topology data errors account for an estimated 15–20% of unplanned outages [64]. By enabling continuous automated topology validation, the framework contributes to reducing outage frequency and duration, with downstream benefits for economic productivity and public safety. The open-source implementation (GitHub repository) lowers the barrier to adoption for utilities with limited software development resources.

8.13 Cost-Benefit Analysis of Recall-Oriented Design

The 44.8% precision may appear low by machine learning standards, but the operational context fundamentally alters the cost function. In power distribution systems, the cost of a missed anomaly (C_{FN}) includes potential cascading failures, equipment damage, and safety hazards, while the cost of a false alarm (C_{FP}) is limited to operator review time. Industry analyses estimate $C_{\text{FN}}/C_{\text{FP}} \geq 50$ for distribution network monitoring [7]. Under this asymmetric cost model, the optimal operating point satisfies Recall ≥ 0.95 even at the expense of Precision ≤ 0.50, which our framework achieves ($R = 78.1\%$, $P = 44.8\%$).

To formalize, the expected cost per anomaly is:

$$\mathbb{E}[C] = C_{\text{FN}} \cdot (1 - R) + C_{\text{FP}} \cdot \frac{1 - P}{P} \cdot R, \tag{18}$$

where R is recall and P is precision. For $C_{\text{FN}}/C_{\text{FP}} = 50$, $R = 0.781$, and $P = 0.448$, the expected cost ratio is $\mathbb{E}[C_{\text{ours}}]/\mathbb{E}[C_{\text{high-P}}] = 0.23$ when compared against a hypothetical high-precision system ($R = 0.60$, $P = 0.90$), confirming that the recall-oriented design minimizes operational risk. The explainability module further reduces C_{FP} by enabling rapid triage: operators can dismiss false positives within seconds when the reasoning chain shows low confidence or inconsistent evidence.

8.14 Limitations

We acknowledge four limitations that contextualize our results.

1. **Three-phase modeling.** The current implementation uses single-phase approximation. Full three-phase unbalanced modeling requires the PandaPower 3-phase extension, which is planned for future work.
2. **Training data.** The GNN layer uses synthetically generated anomaly data. While synthetic injection is a standard practice, validation on real-world SCADA data with naturally occurring anomalies remains essential.
3. **Real-time performance.** Although average processing times are practical (0.94 s, ranging from 0.16 to 1.13 s), achieving sub-second real-time monitoring for networks with >1 000 buses requires further optimization through parallel partition processing or GPU-accelerated computation.
4. **Measurement redundancy.** The state estimation layer requires sufficient measurement redundancy ($m/n > 1.5$). In low-observability regions, detection capability is limited.

Despite these limitations, the framework provides a practical and extensible foundation for topology data quality assurance. The modular architecture allows each limitation to be addressed independently—for example, replacing the single-phase SE module with a three-phase variant does not affect the rule engine or GNN layers.

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Competing Interests

The authors declare that they have no competing interests.

9 Conclusion

This paper addresses a fundamental challenge at the intersection of computer science and power systems engineering: how to reconcile heterogeneous data representations—graph topology, time-series telemetry, and symbolic engineering models—under physical constraints for reliable infrastructure operation. We make three contributions to this end.

First, we demonstrate that no single algorithmic paradigm suffices for comprehensive topology data quality assurance. Rule-based reasoning detects structural anomalies but misses measurement-level deviations; physics-based state estimation identifies measurement anomalies but requires sufficient observability; GNN-based detection learns complex patterns but lacks interpretability. The three-layer hybrid architecture unifies these paradigms, achieving 78.1% global anomaly recall (83.5% with improved GNN) across 30 networks spanning 3 to 1 888 buses—a 3.2× improvement over statistical baselines—while maintaining 0.94 s average processing time.

Second, we show that standard benchmarks are insufficient for evaluating hybrid architectures. On the standard benchmark, removing any single layer has zero impact on recall (RQ2), potentially misleading practitioners into concluding that the layers are redundant. A dedicated hard anomaly experiment reveals the true complementary coverage: removing SE drops recall by 40 pp, removing GNN by 23 pp, and removing Rule by 7 pp (RQ3). This finding has implications beyond power systems: *hybrid AI systems should be evaluated with component-specific benchmarks that exercise each layer's unique capabilities*.

Third, we demonstrate that explainability is not merely an optional feature but a prerequisite for operator adoption in safety-critical domains. The integrated explainability module—providing reasoning chains and perturbation-based feature importance—enables operators to triage false positives within seconds, effectively reducing the operational cost of the recall-oriented design. The cost-benefit analysis (Section 8.12) formalizes this trade-off, showing that the recall-oriented configuration reduces expected operational cost by 77% compared to a hypothetical high-precision system.

Four directions warrant investigation: (i) three-phase unbalanced state estimation for heavily unbalanced feeders; (ii) validation on real-world SCADA data with naturally occurring anomalies—the 46.8% recall on extended anomaly types (RQ7) indicates significant room for improvement through real-world training; (iii) harder anomaly benchmarks that exercise cross-layer complementary detection; and (iv) parallel partition processing for real-time monitoring of networks exceeding 1 000 buses. The open-source implementation and benchmark datasets are available at `https://github.com/secret-justice/topology-anomaly-detection`.

Data Availability

The experimental data, including all benchmark results (JSON), trained GNN models (.pt), and visualization outputs, are available at `https://github.com/secret-justice/topology-anomaly-detection`. All experiments are reproducible with fixed random seeds (42–46) and documented hyperparameters. The test networks use publicly available datasets: PandaPower [58], SimBench [57], CIGRE [56], and IEEE standard feeders [54].

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